

CAS Machine Learning – Time Series

Base concepts

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Informatik



Outline

- What is a time series?
- Time series patterns
- Visualization of time series
- Data generation process
- Modeling time series
- Auto-correlation, white noise
- Signal + noise model
- Random walks
- Time series decomposition

What is a time series?

A time series is a collection of **data points** observed **sequentially in time**.

A time series can be

- Continuous: continuous observations over a time interval e.g., analog audio signal
- **Discrete**: data collected at distinct & separate time points e.g., digital audio signal

Interval between data points can be

- Irregular e.g., patient data collected during medical appointments
- **Regular** e.g., data collected every hours, months

Time interpretation is secondary, important is for the **indexing variable** to be **ordered**.

Time series examples – Finance

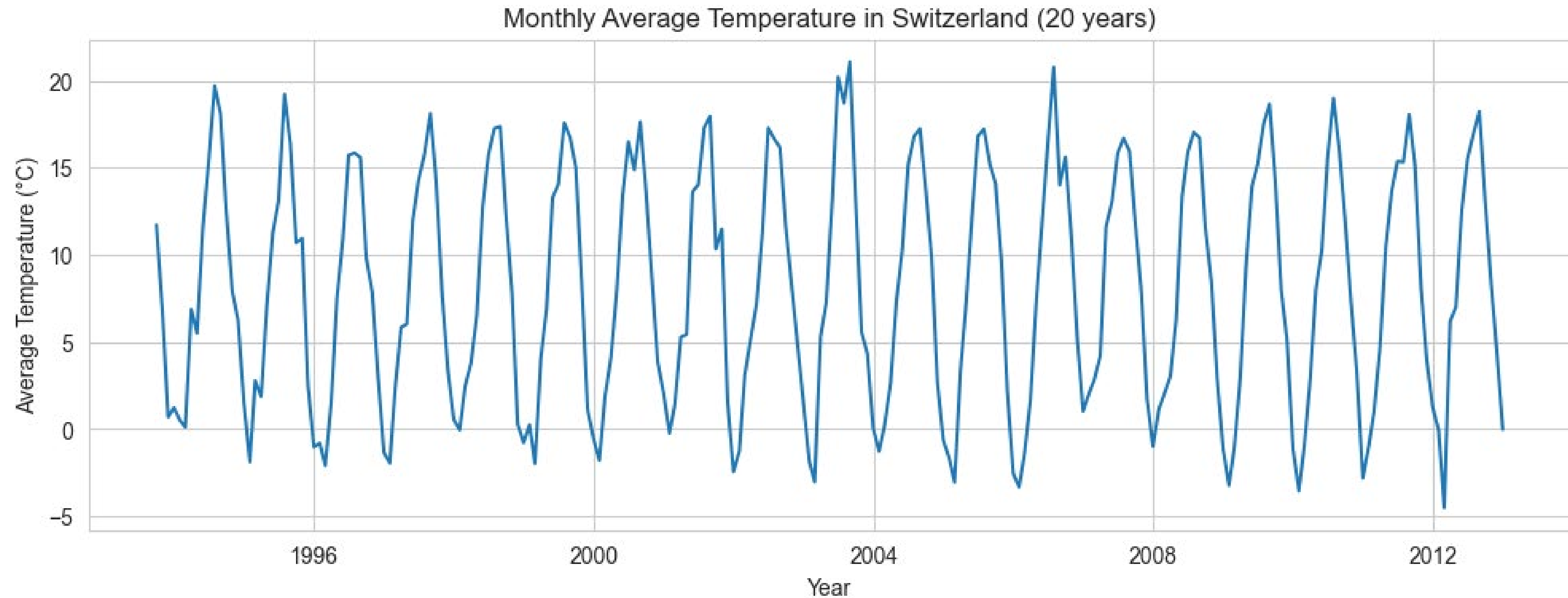


Time series patterns – Trend

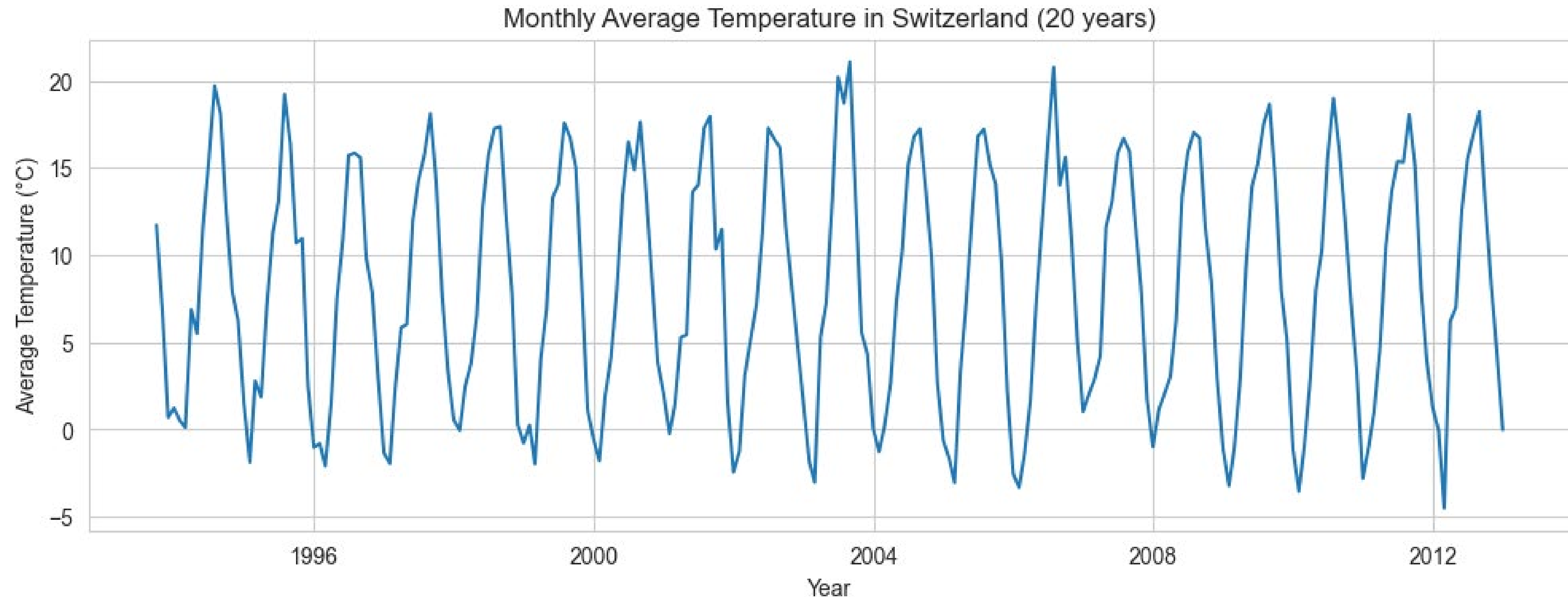


A **trend** is a long-term movement or direction in a time series that shows a consistent upward or downward trajectory over time, i.e., a long-term change in the mean.

Time series examples – Meteorology

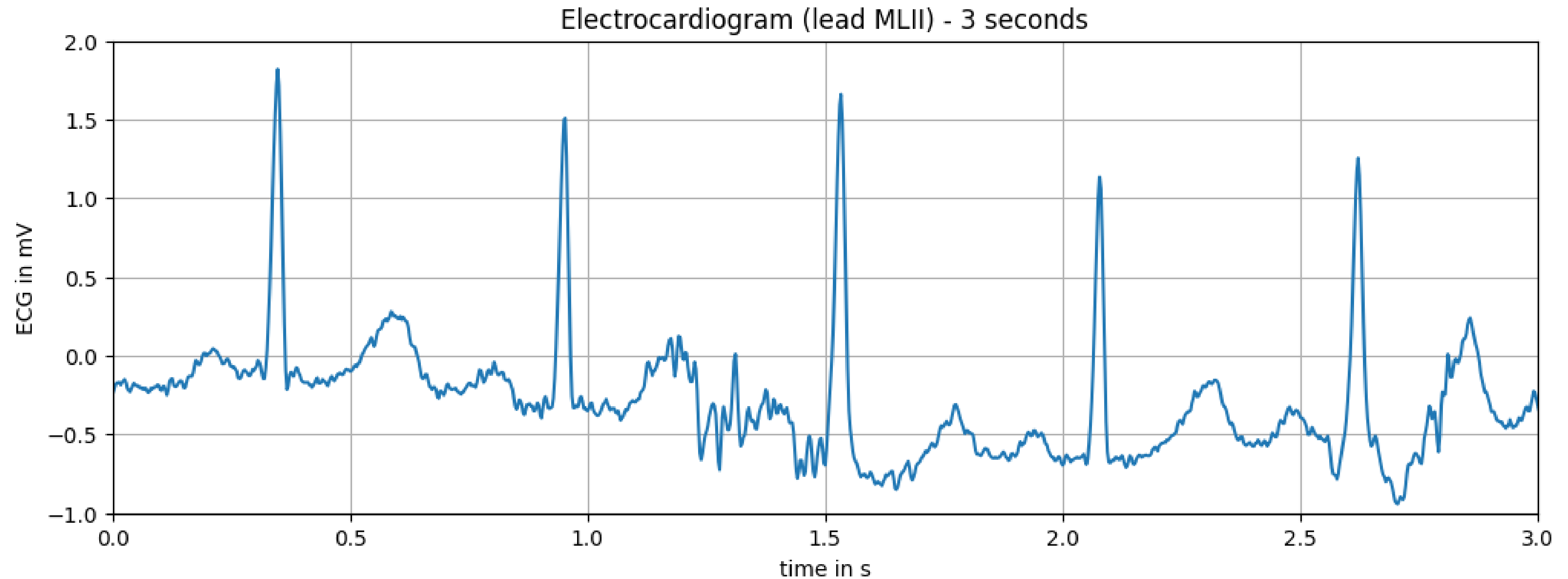


Time series patterns – Seasonal

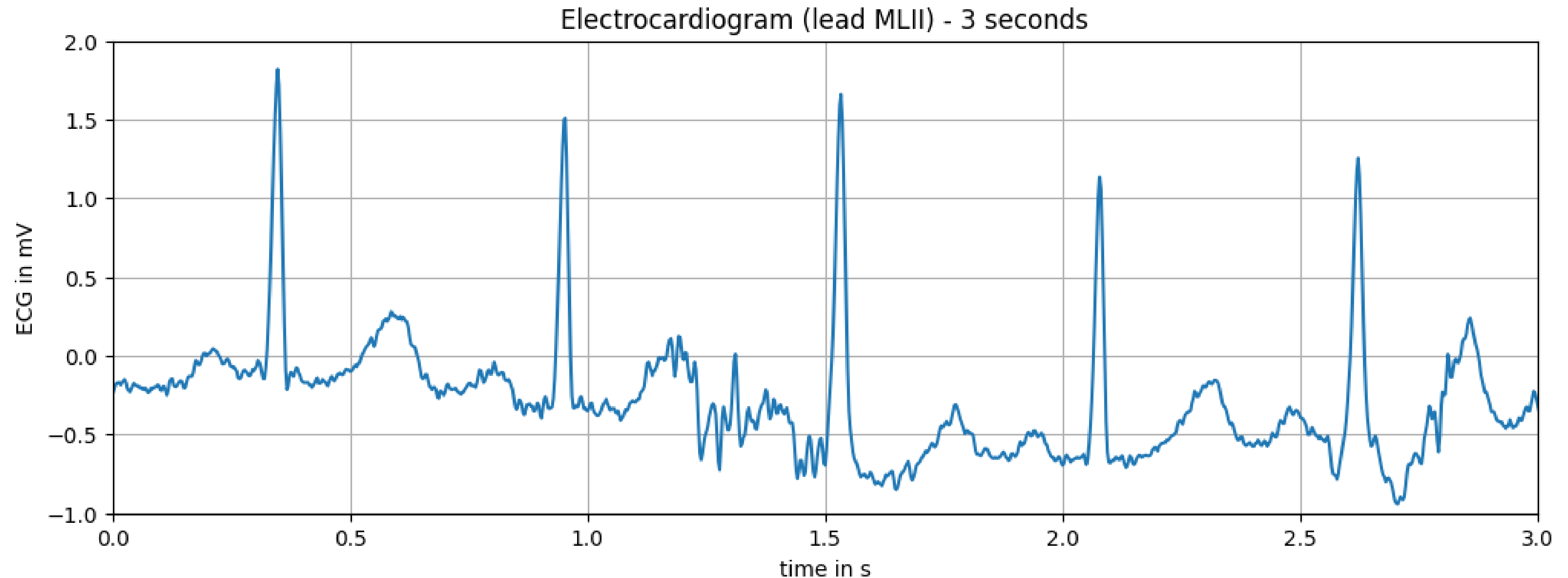


A **seasonal** behavior is a repeating pattern with fixed period, typically tied to the calendar, such as higher/lower temperatures in summer/winter or increase in retail sales at year-end.

Time series examples – Healthcare

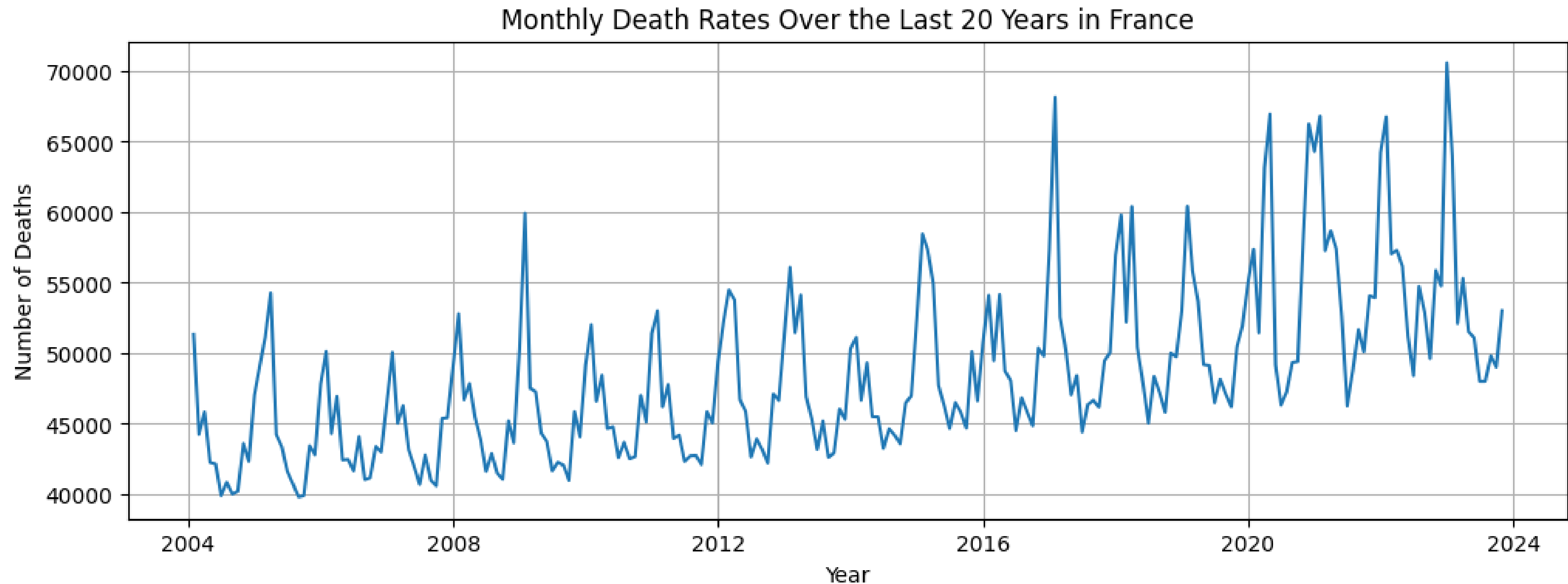


Time series patterns – Cycle

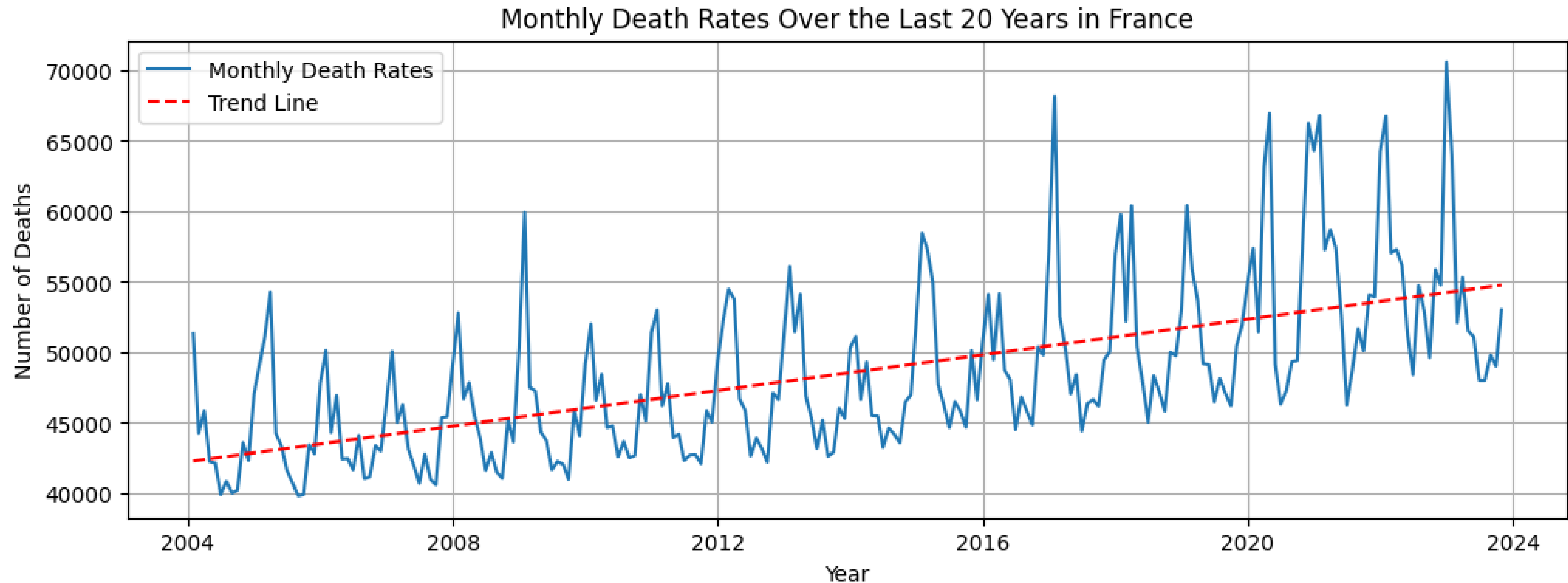


A **cyclic** behavior is a pattern that repeats over regular or irregular intervals, such as heartbeats in an ECG or economic cycles of expansion and recession.

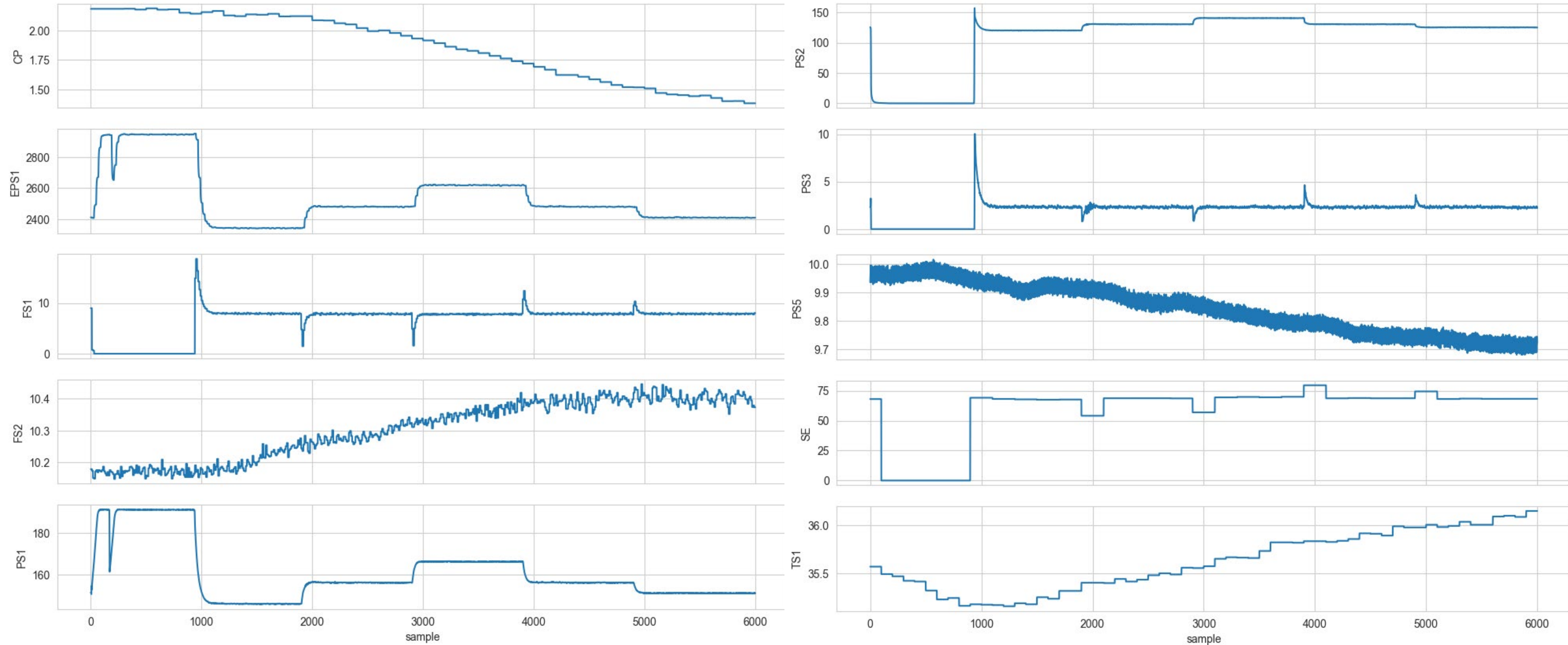
Time series examples – Demographics



Time series patterns – Trend + Seasonal



Time series examples – Industry



Note on periodic signals and aliasing

Consider the periodic signal $x_t = \cos(2\pi f \Delta t)$ sampled at interval Δ i.e., x_t is observed at time Δt .

Increasing f from 0, the **fastest observable oscillation** is reached when $f = \frac{1}{2\Delta}$ (called the **Nyquist frequency**)

- In this example, $x_t = \cos(\pi t) = (-1)^t$.

Higher frequencies are folded down into the interval $\left[0, \frac{1}{2\Delta}\right]$: taking f such that $\frac{1}{2\Delta} < f < \frac{1}{\Delta}$, and $f' = \frac{1}{\Delta} - f$, we get

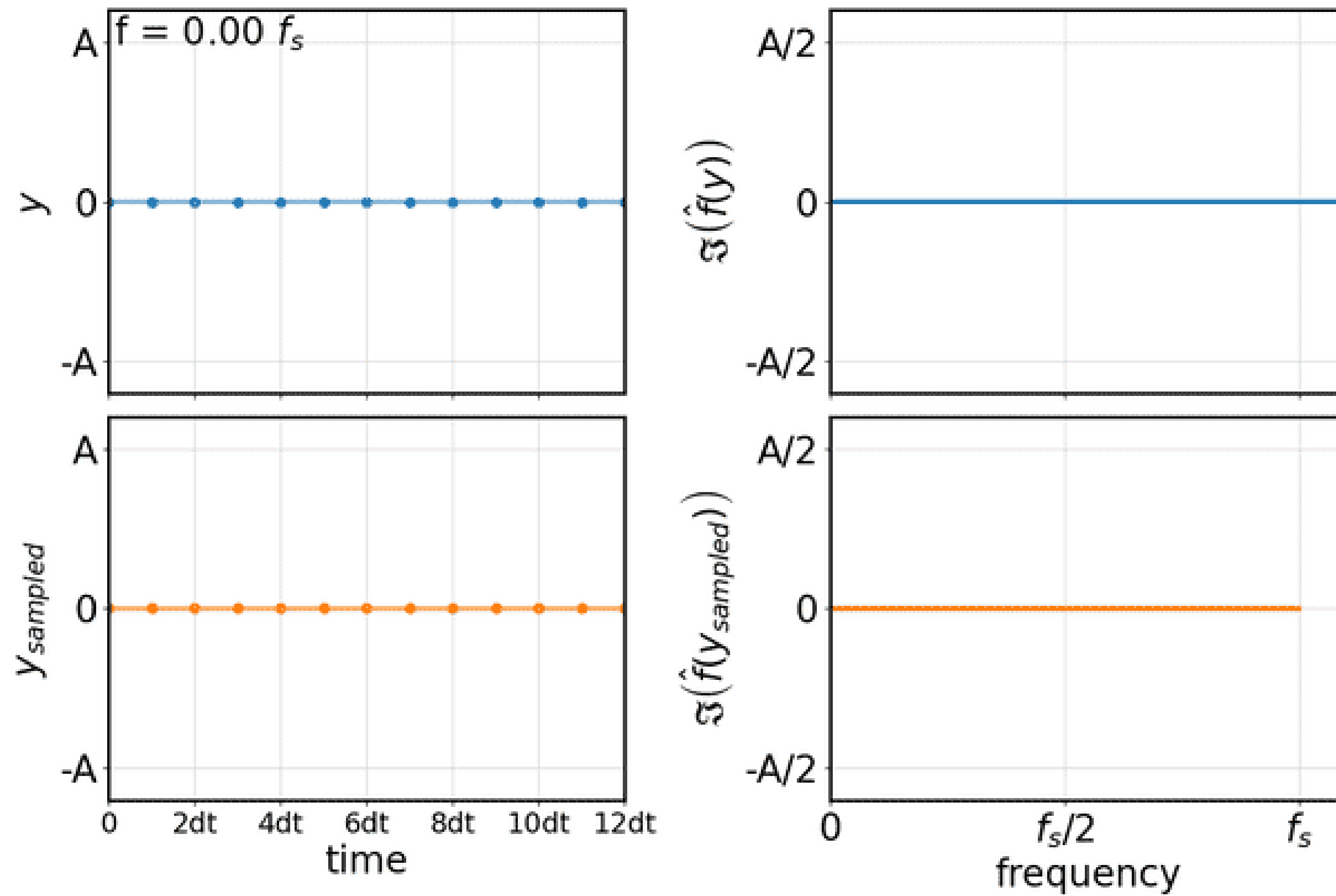
$$x_t = \cos(2\pi f \Delta t) = \cos\left(2\pi\left(\frac{1}{\Delta} - f'\right)\Delta t\right) = \cos(2\pi(t - f'\Delta t)) = \cos(2\pi f' \Delta t)$$

Then f and f' are indistinguishable, hence **aliases** of each other.

- For arbitrary f the observed frequency is $f' = |f - k\Delta|$ with k chosen such that $f' \leq \frac{1}{2\Delta}$

To capture a signal's frequency without aliasing, the sampling rate must be at least **twice** the signal's frequency (**Nyquist-Shannon sampling theorem**).

Note on periodic signals and aliasing



Davidjessop, wikimedia

Visualization – Preparing the data

Review data structure

- Ensure **consistent** time intervals
- Granularity of observations: up/down-sampling

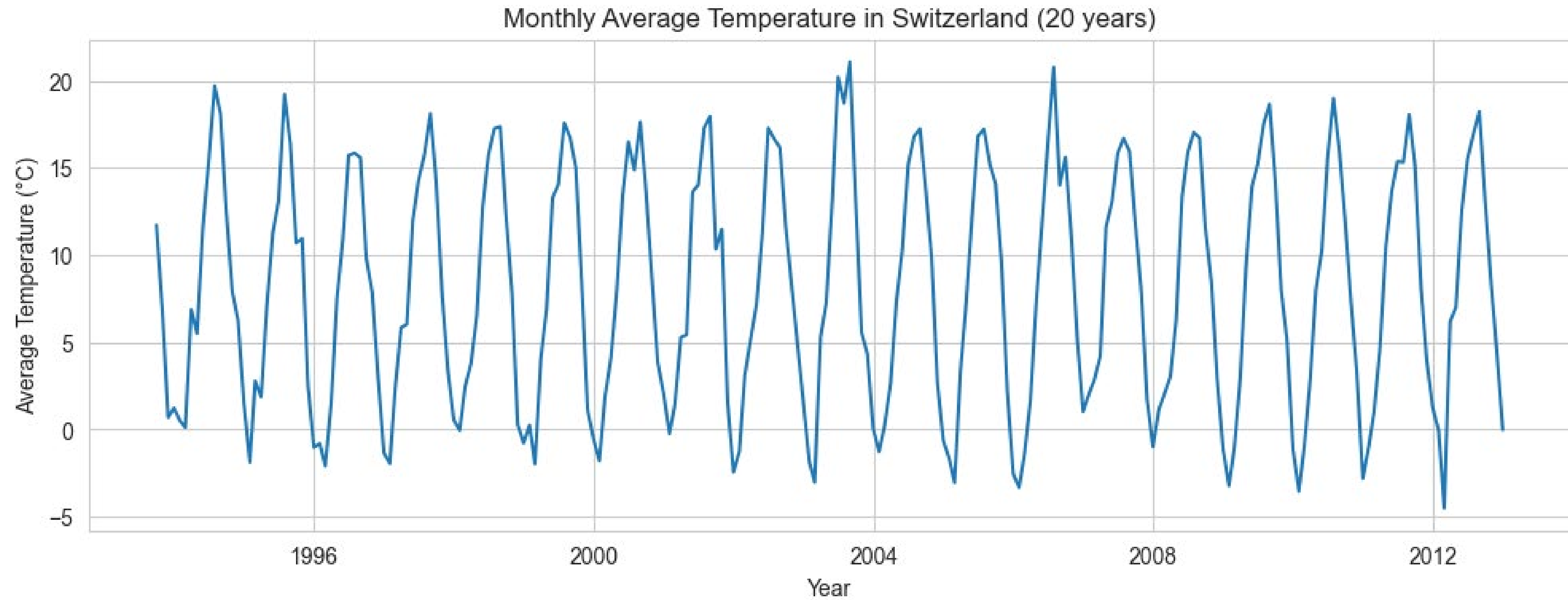
With **missing data**, start by asking the following questions

- What is the underlying **reason** for the missing data?
- Is the data **missing at random or is there a pattern?**
- How much data is missing, and could imputing it introduce **bias** or distort the analysis?
- Would **removing** the missing data be more appropriate in the context?

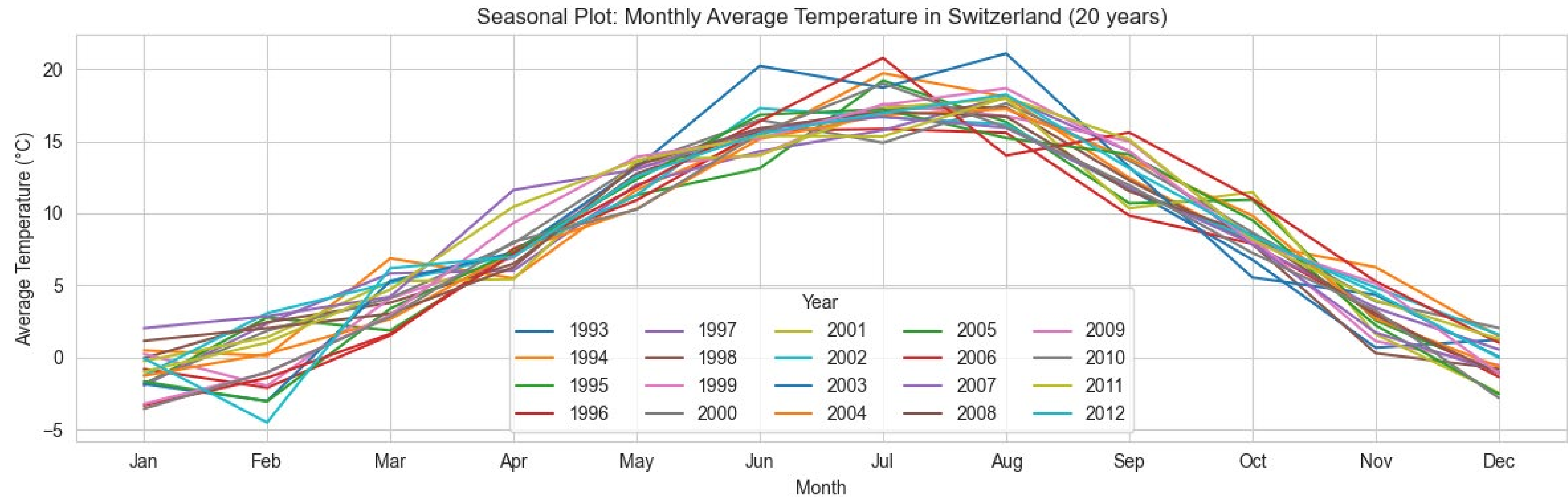
Based on the answers decide whether to proceed with **imputation**

- Filling: forward, backward, mean
- Interpolation: nearest, linear, polynomial, seasonal

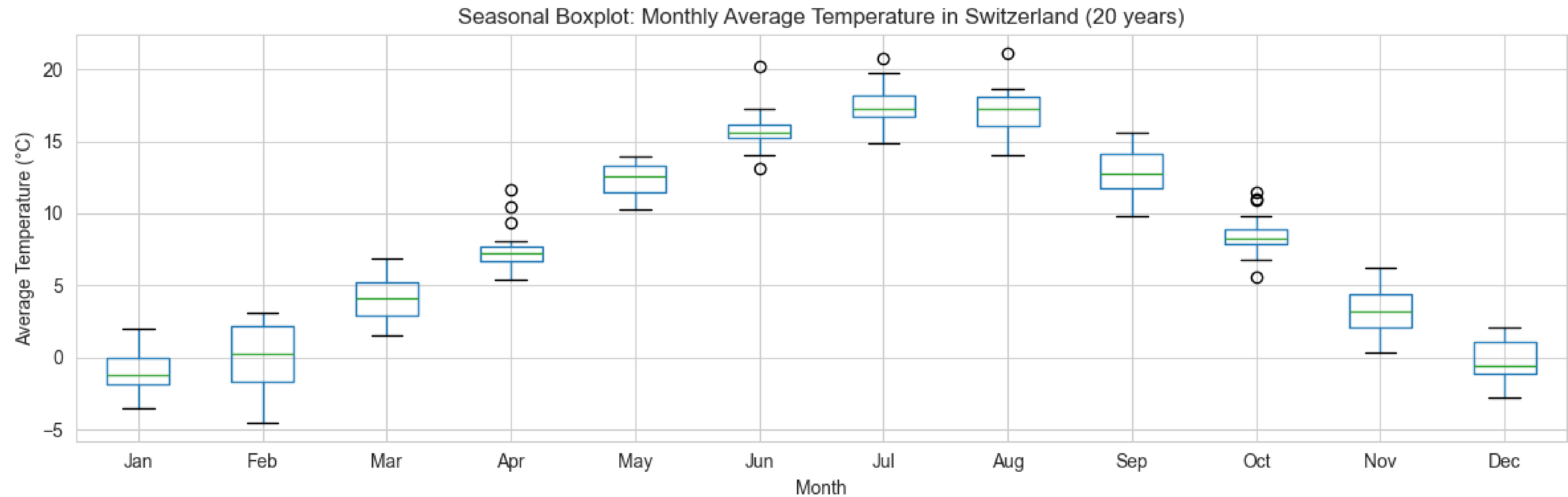
Visualization – Time plot



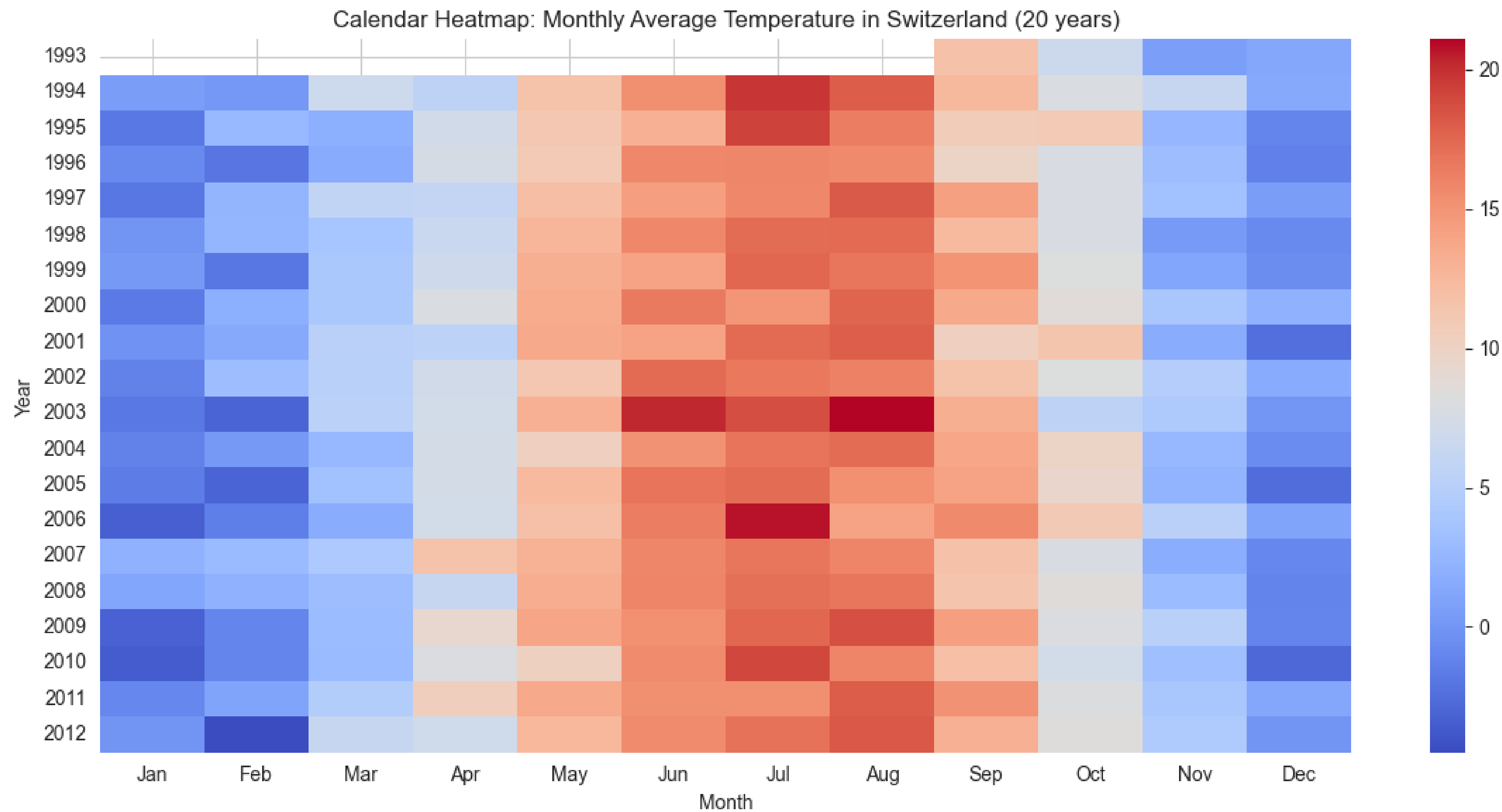
Visualization – Seasonal plot



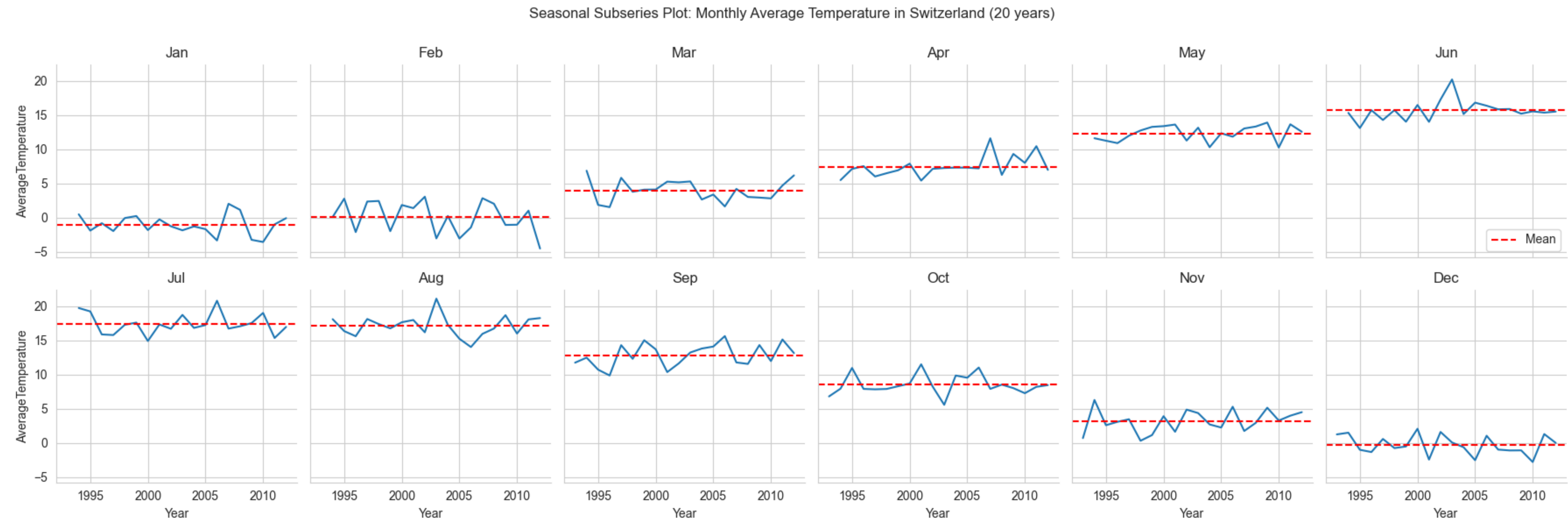
Visualization – Seasonal boxplot



Visualization – Heatmap

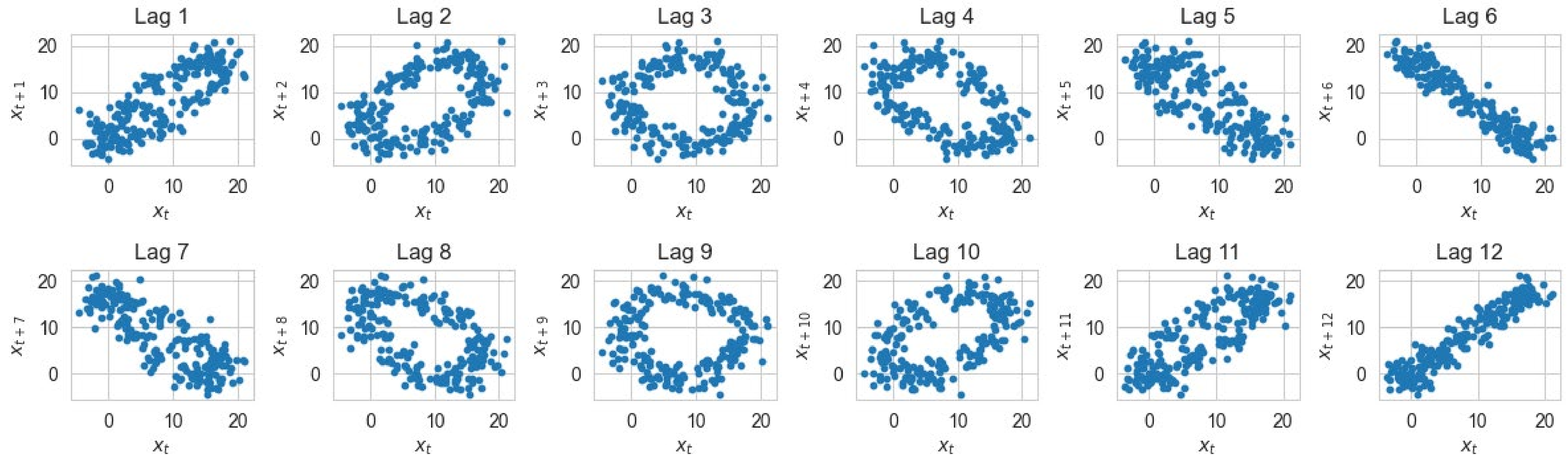


Visualization – Seasonal subseries plot



Visualization – Lag plot

Lag Plot: Monthly Average Temperature in Switzerland (20 years)



Data generating process (DGP)

The generation of time series data is influenced by:

Deterministic components: known factors, laws, predictable patterns influencing data

- Predictor variables e.g., moon distance for tidal variations
- Trend e.g., long-term growth or decline
- Seasonal variations e.g., holiday season, weather patterns
- Cyclical patterns e.g., business cycles, tidal patterns

Stochastic components: random fluctuations introducing uncertainty

- Noise e.g., random errors in measurement or observation
- Irregular events e.g., natural disasters, economic crises

DGP example – Electrocardiogram (ECG)

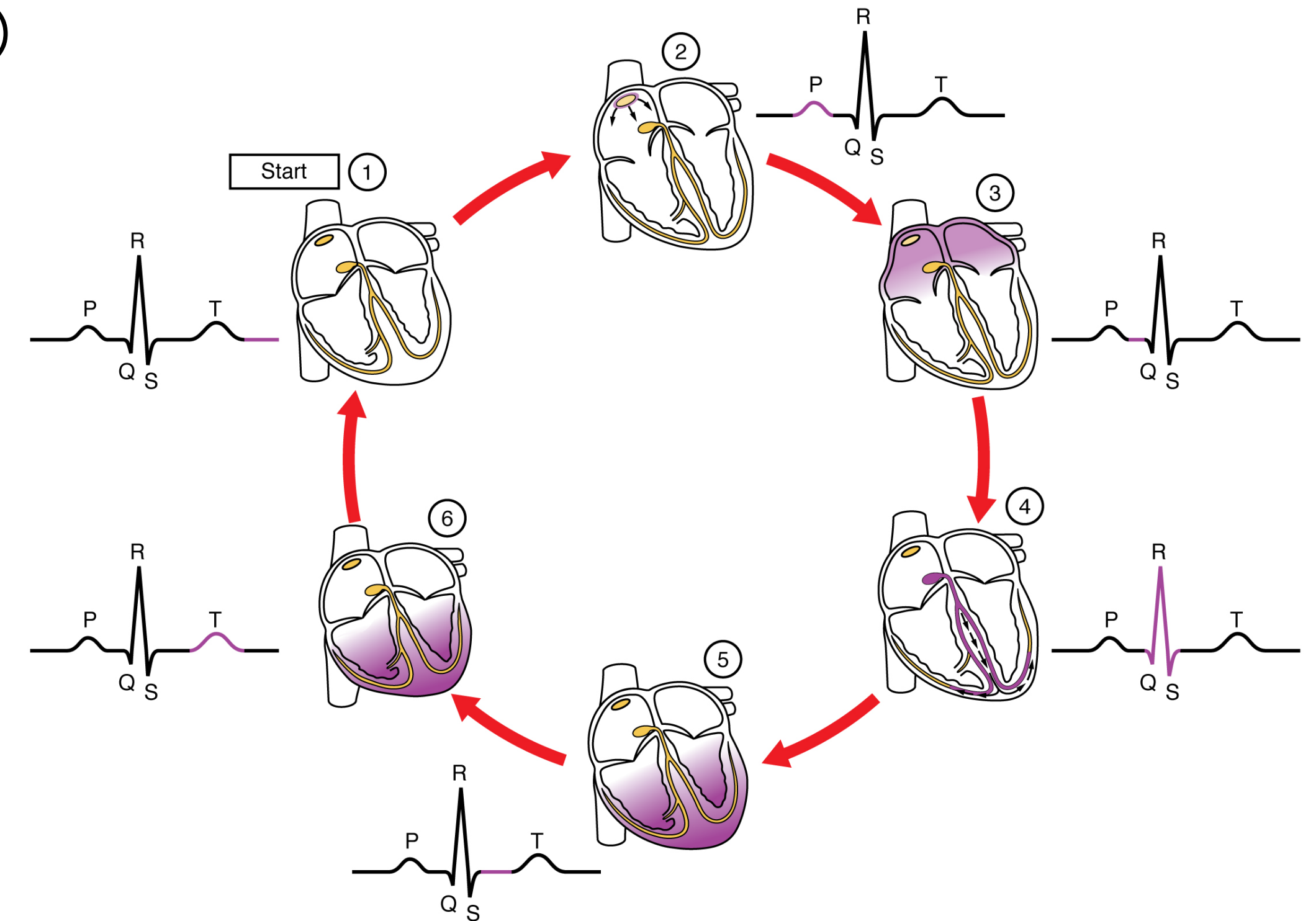
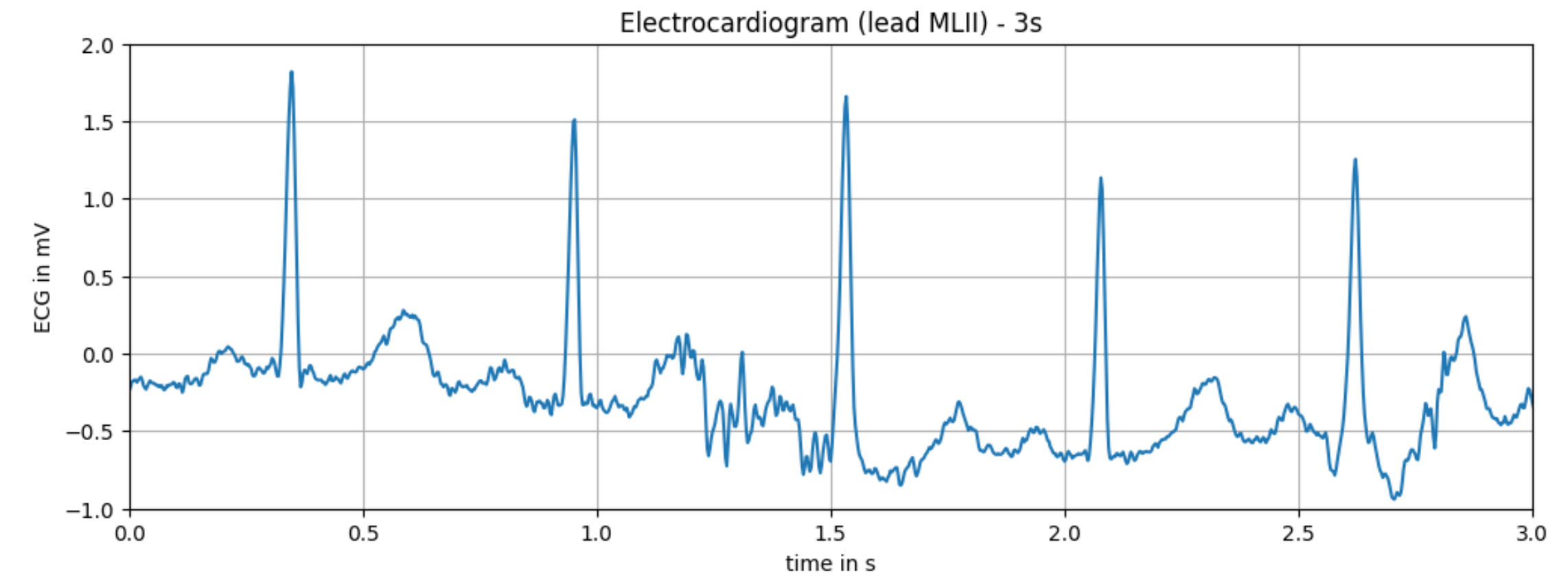
Deterministic components

- Cardiac cycle (P wave, QRS complex, and T wave)
- Heart rate (based on physiological state e.g., sleep, exercise)

Stochastic components

- Electrical noise from the recording equipment
- Movement artifacts from patient motion
- Irregular heartbeats or arrhythmias

Modeling ECG data requires approaches that account for **both** the deterministic and stochastic components.



OpenStax College, Wikimedia

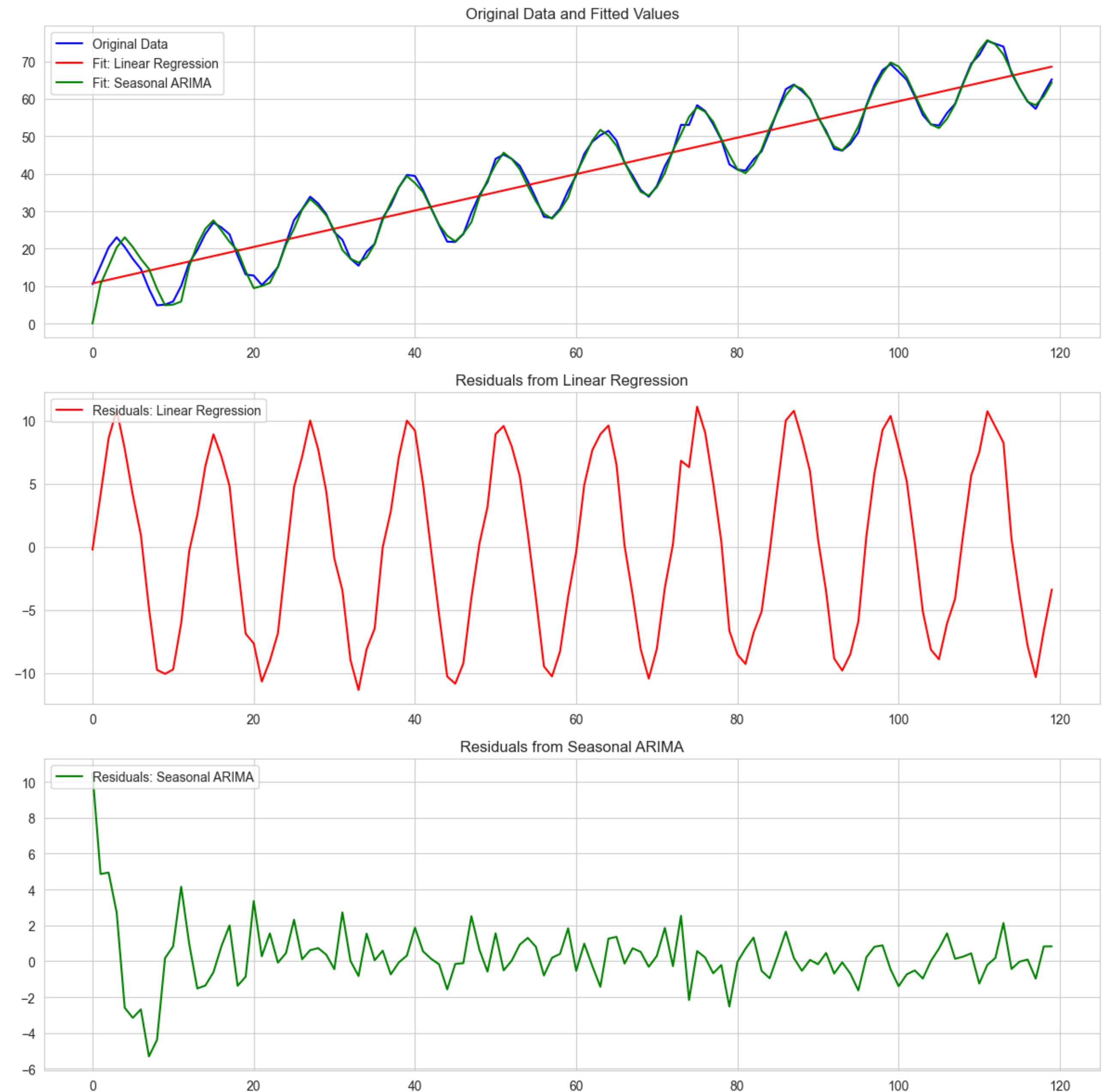
The challenge of time dependence

Traditional statistical methods often assume data points are **independent** and **identically distributed** (IID).

Time series data violates this assumption due to the **correlation between consecutive observations**.

Ignoring this dependence structure can lead to misleading inference and models with poor data fit.

Time series analysis provides a specialized conceptual framework to address these challenges.



Modeling time series

- A. **Understand DGP** i.e., underlying mechanisms/factors that generate/influence the data
- B. Collect a **sufficient** amount of **representative** data (dependent on modeling objective and data)

In practice, make simplifying **assumptions** (based on the modeling objective and data) such as:

- **Stationarity**: The statistical properties of the data remain constant over time.
- **Linearity**: The relationship between variables is linear.
- **Normality**: The data follows a normal distribution.
- **Decomposability**: The time series can be broken down into trend, seasonal, and irregular components.

Modeling objectives:

- **Forecasting**: prioritize trends and seasonal patterns while downplaying stochastic variations
- **Classification**: assume extracted patterns and features can differentiate between categories
- **Anomaly detection**: capture deviations from assumed “normal” patterns

Modeling approaches

Time domain approaches: model correlations between adjacent observations

- Directly interpretable in the context of time
- Well-suited for small to moderate datasets.
- E.g., autoregressive integrated moving average, state-space models, Kalman filters

Frequency domain approaches: model periodicity of the time series

- Effective for handling cycles and seasonal effects
- Useful in signal processing
- E.g., Fourier analysis, spectral analysis

Machine learning and deep learning approaches: learn statistical patterns from data

- Model complex nonlinear relationships and patterns
- Require fewer assumptions about data
- E.g., random forest, recurrent/convolutional neural networks

Exercise

Select 2-3 time series datasets: opendata.swiss, ec.europa.eu/Eurostat, Kaggle, etc.

Reflect on the data generation process:

- How was the data collected? Time interval, time range, etc.
- Which factors/laws influence the data?
- Which patterns do you expect to find?

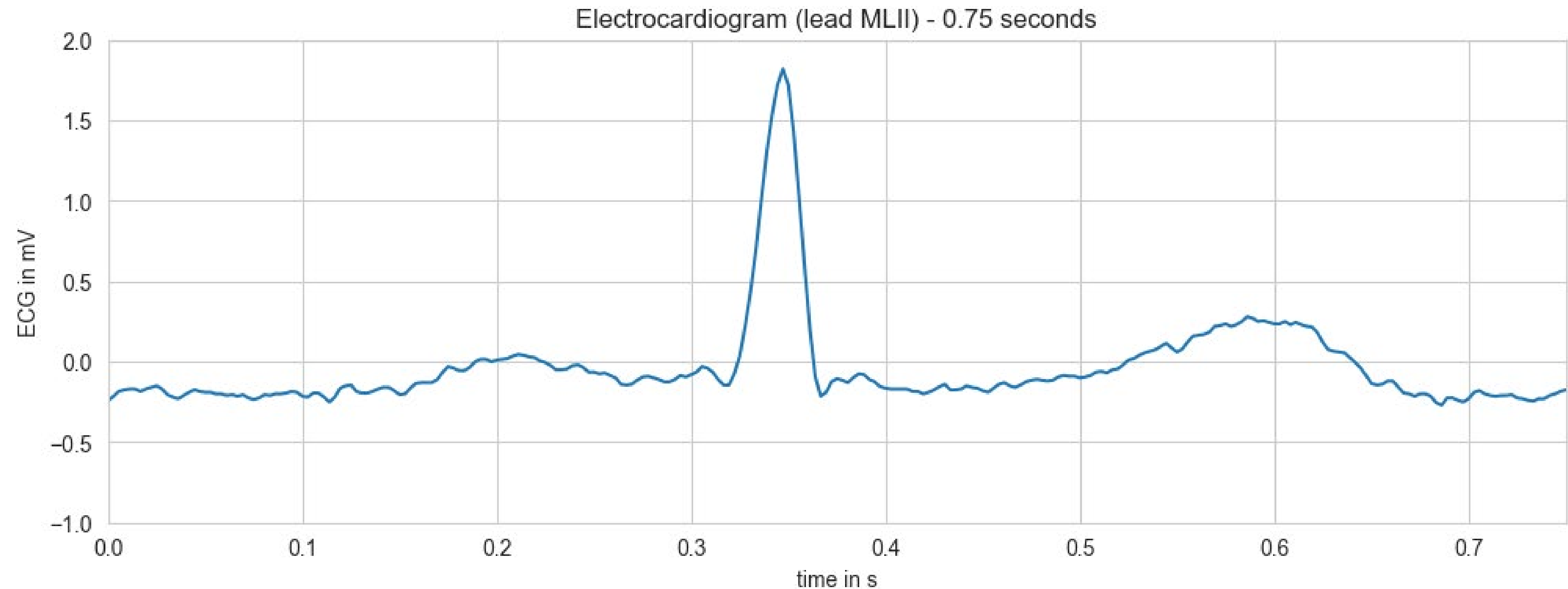
Review notebooks of Modern Time Series Forecasting with Python – Chapter 2 (GitHub repository)

Load and **visualize** datasets with Python.

Analysis:

- How noisy is the data?
- Review patterns, do they match your expectations given your understanding of the DGP?
- Review outliers, can you identify corresponding real-world events?

Time series examples – ECG signal



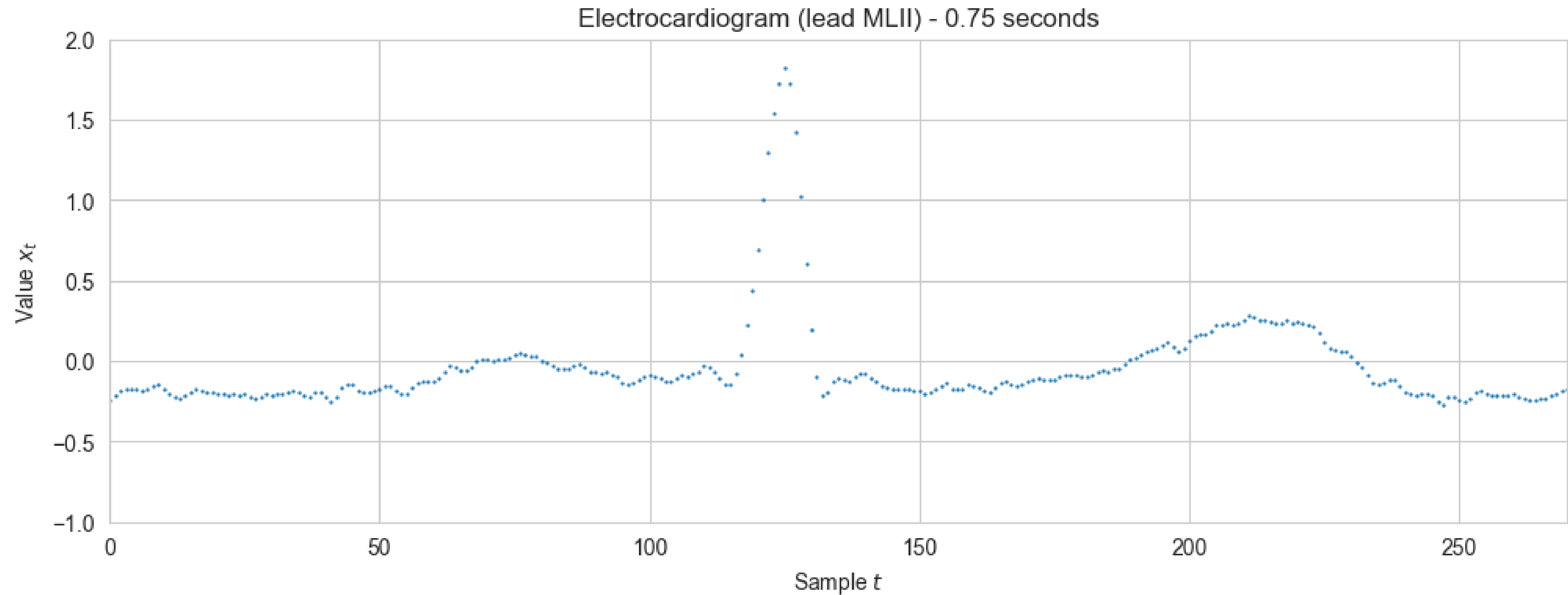
Statistical model for time series

A time series (TS) is a collection of **data points** observed **sequentially in time**.

Model TS as **stochastic processes**, i.e., collections of **random variables** (RV), $\{X_1, X_2, \dots, X_n\}$, indexed according to their observation order.

A model specifies the **joint distribution** of the sequence of RVs $P[X_1 \leq x_1, \dots, X_n \leq x_n]$ in the continuous case, $P[X_1 = x_1, \dots, X_n = x_n]$ in the discrete case, where $\{x_1, x_2, \dots, x_n\}$ is a **realization** of the stochastic process.

A realization of a stochastic process



$ecg: [-0.245, \quad -0.215, \quad -0.185, \quad -0.175, \quad -0.17, \quad -0.17, \quad -0.185, \quad -0.17, \quad -0.16, \quad -0.15, \quad \dots]$

$ecg: [x_0, \quad x_1, \quad x_2, \quad x_3, \quad x_4, \quad x_5, \quad x_6, \quad x_7, \quad x_8, \quad x_9, \quad x_{10}, \quad \dots]$

Recap – Probability concepts

Random variables (RV) are **real-valued functions** whose outcomes vary due to ... randomness.

Property	Expectation	Variance	Covariance
Intuition	Long-term average, mean	Spread around mean	Joint variability, strength and direction of linear relationship
Formula	$E[X] = \sum_x x \cdot P(X = x) = \mu_X$	$Var(X) = E[(X - \mu_X)^2] = \sigma_X^2$	$cov(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = \gamma_{X,Y}$
Linearity	$E[aX + b] = aE[X] + b$	$Var(aX + b) = a^2Var(X)$	$cov(aX + b, cY + d) = ac \cdot cov(X, Y)$
Additivity	$E[X + Y] = E[X] + E[Y]$	$Var(X + Y) = Var(X) + Var(Y) + 2cov(X, Y)$	$cov(X + Y, Z) = cov(X, Z) + cov(Y, Z)$

Note that $\gamma_{X,Y} = \gamma_{Y,X}$ (**symmetric**), that $\gamma_{X,Y} = E[XY] - \mu_X\mu_Y$ and that $\gamma_{X,X} = Var(X)$

We say that X and Y are **independent** $\Leftrightarrow P(X = x, Y = y) = P(X = x)P(Y = y) \Rightarrow \gamma_{X,Y} = 0$

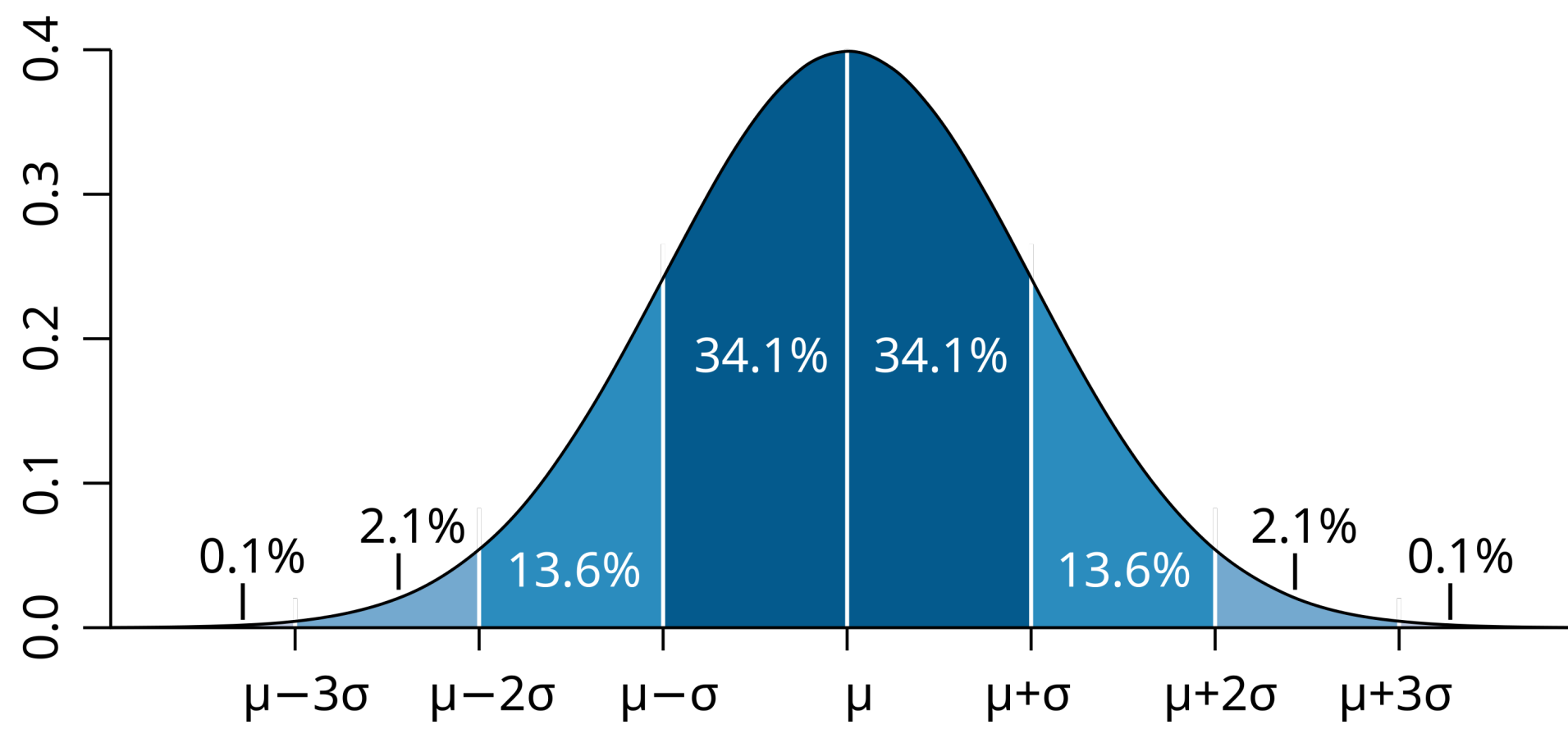
The **correlation** is the normalized covariance: $cor(X, Y) = \frac{\gamma_{X,Y}}{\sigma_X\sigma_Y} = \rho_{X,Y}$ with $-1 \leq \rho_{X,Y} \leq 1$

With TS, the **auto-covariance** is denoted $\gamma_{X_s,X_t} = \gamma(s, t)$ and the **auto-correlation** $\rho_{X_s,X_t} = \rho(s, t)$

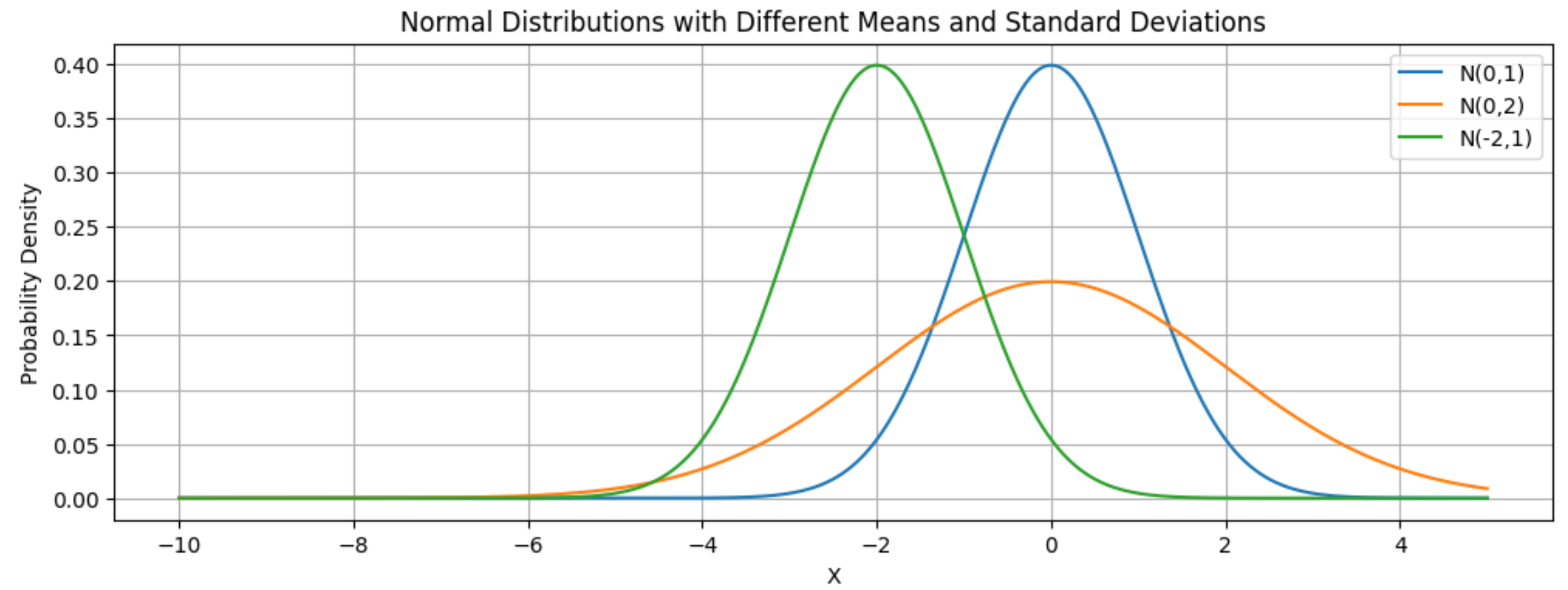
Recap – Probability concepts

Probability distributions describe how probabilities are distributed over the values of the random variables.

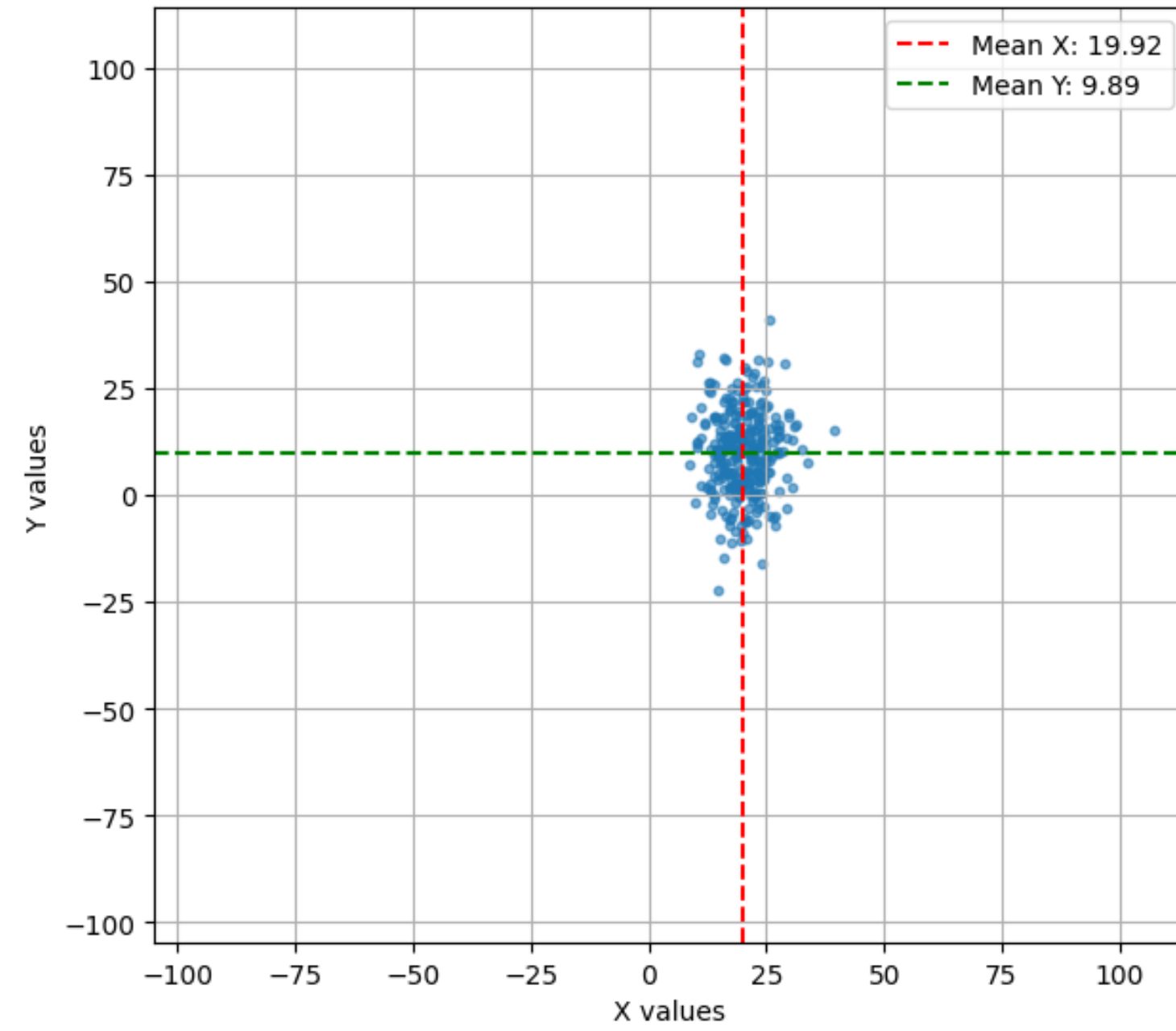
Normal distribution $\mathcal{N}(\mu, \sigma^2)$: $f(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$



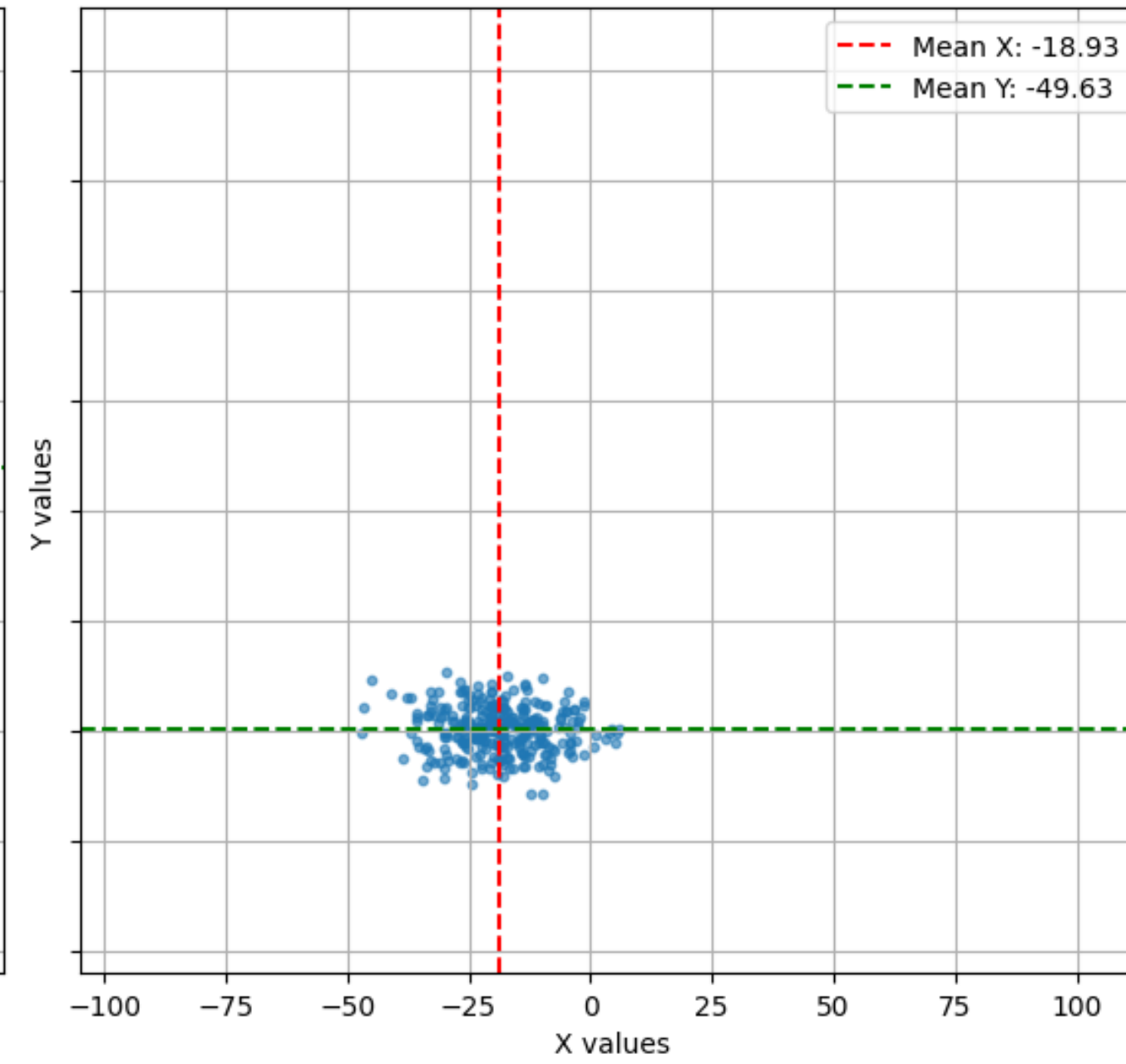
Ainali, Wikipedia



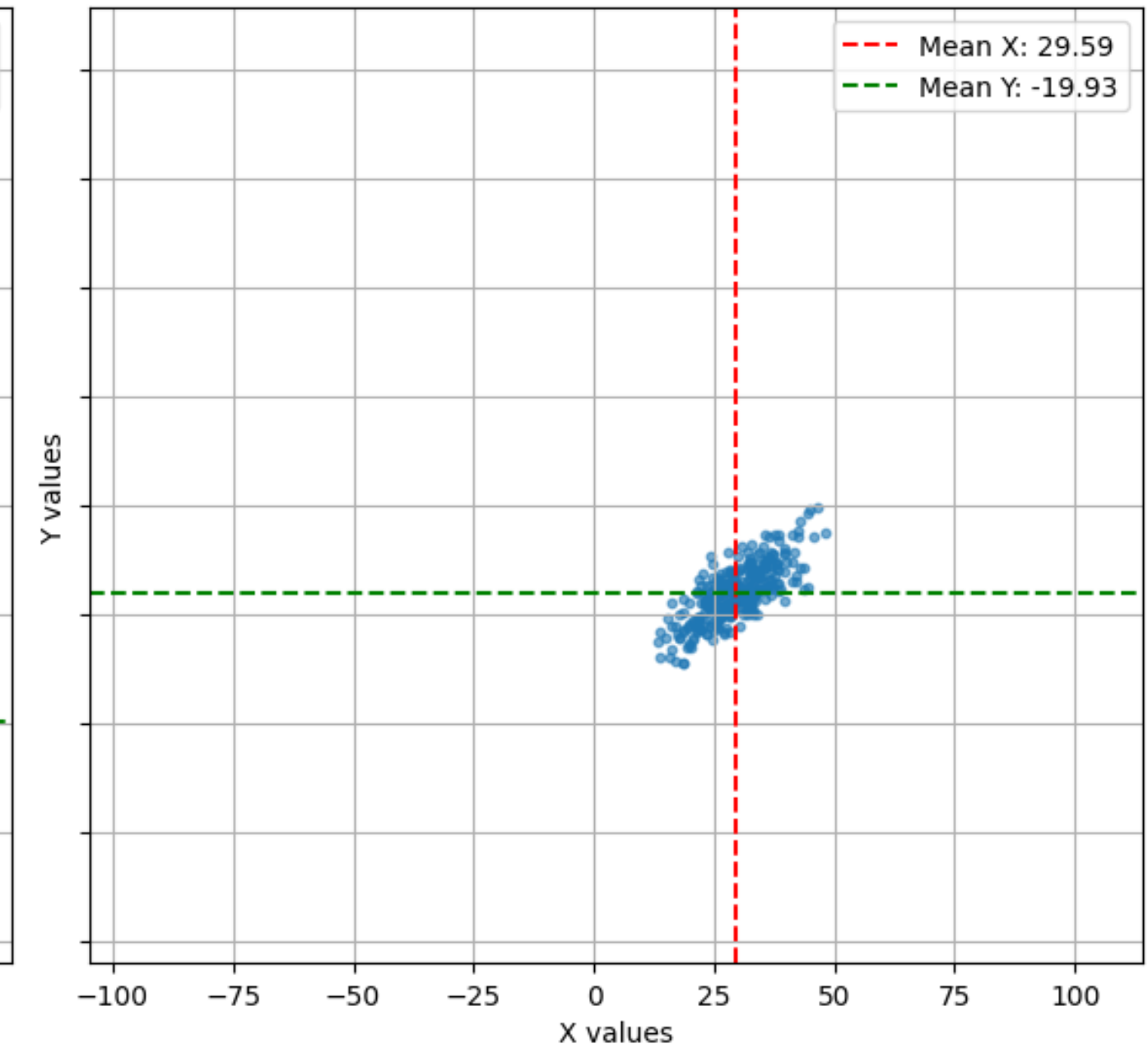
Low Var X, High Var Y, No Corr
Covariance: 0.64



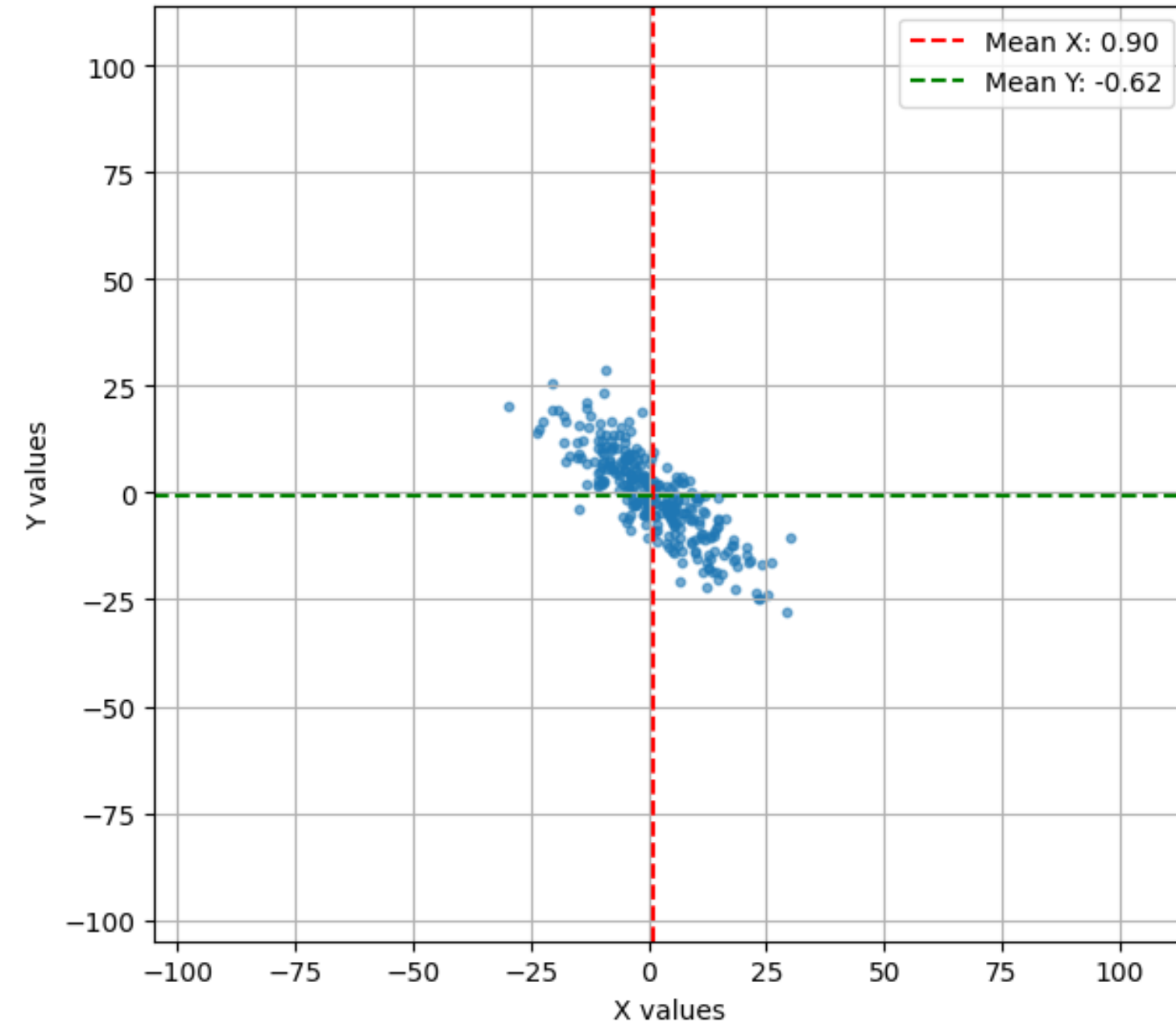
High Var X, Low Var Y, No Corr
Covariance: -4.41



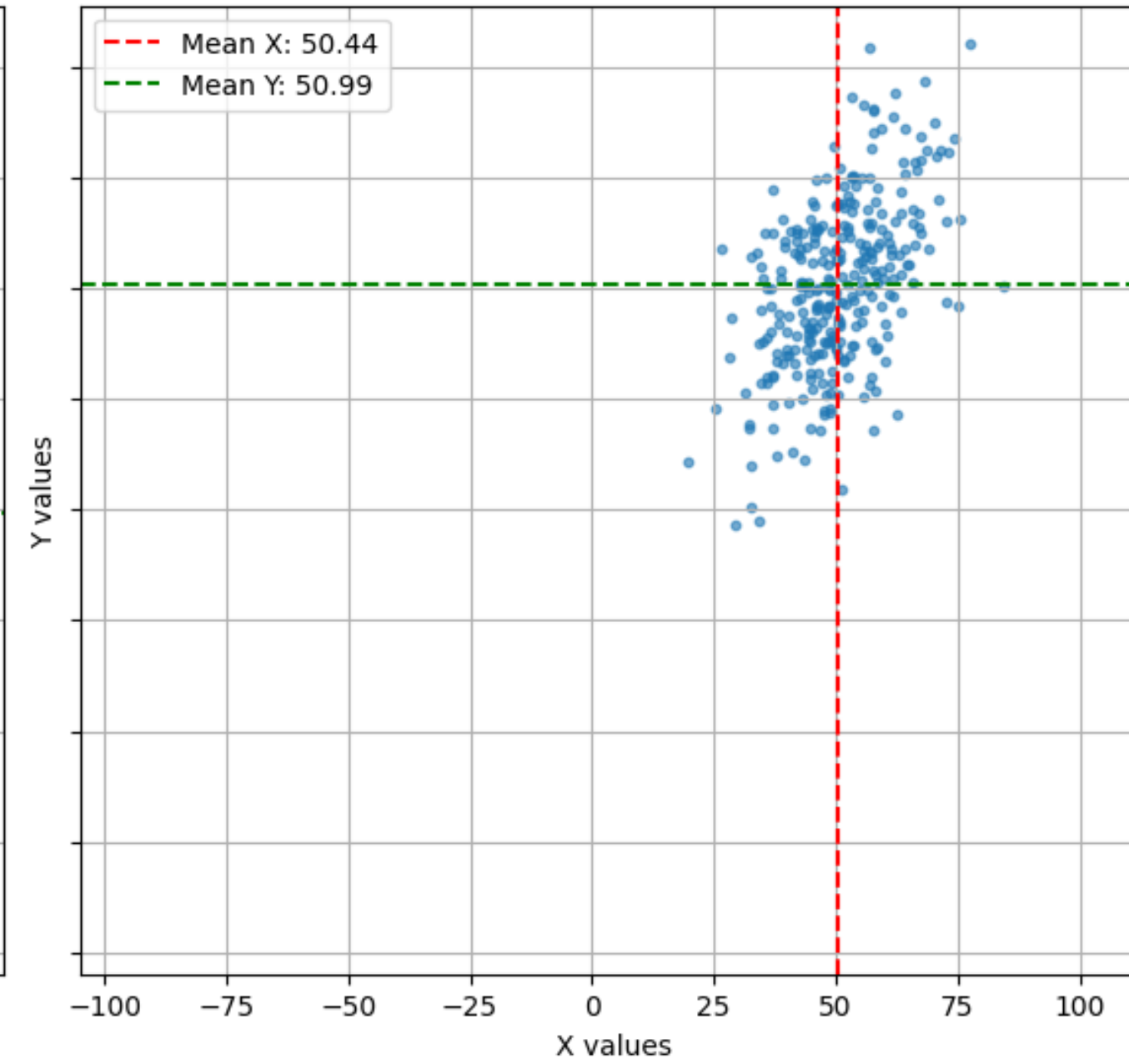
Same Var, High Corr
Covariance: 33.21



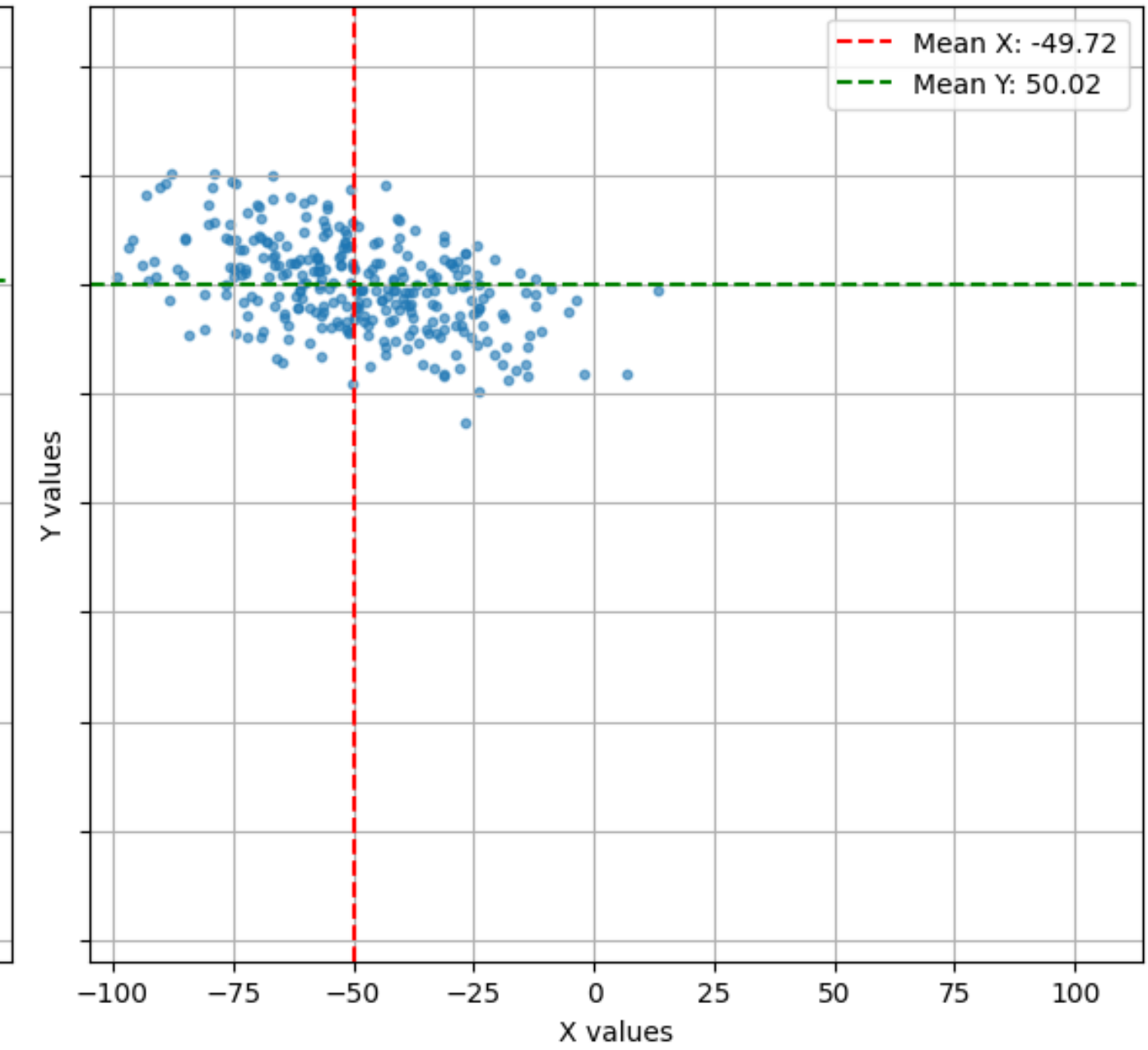
Same Var, High Neg Corr
Covariance: -84.92



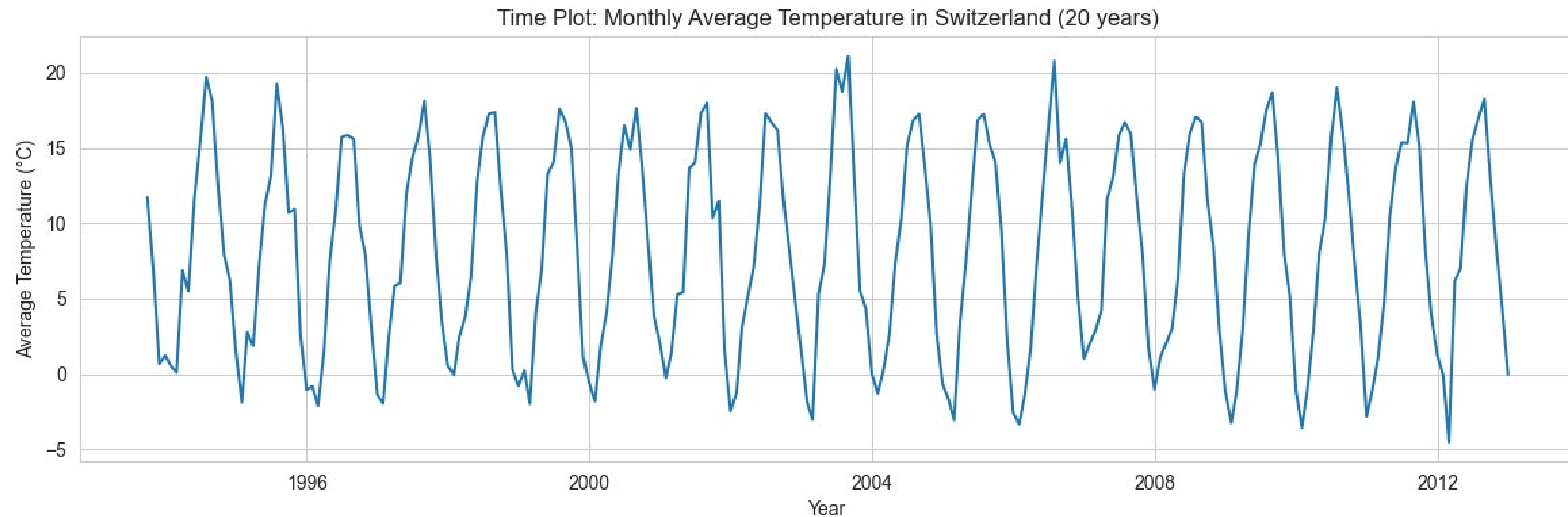
High Var X, Higher Var Y, Moderate Corr
Covariance: 100.42



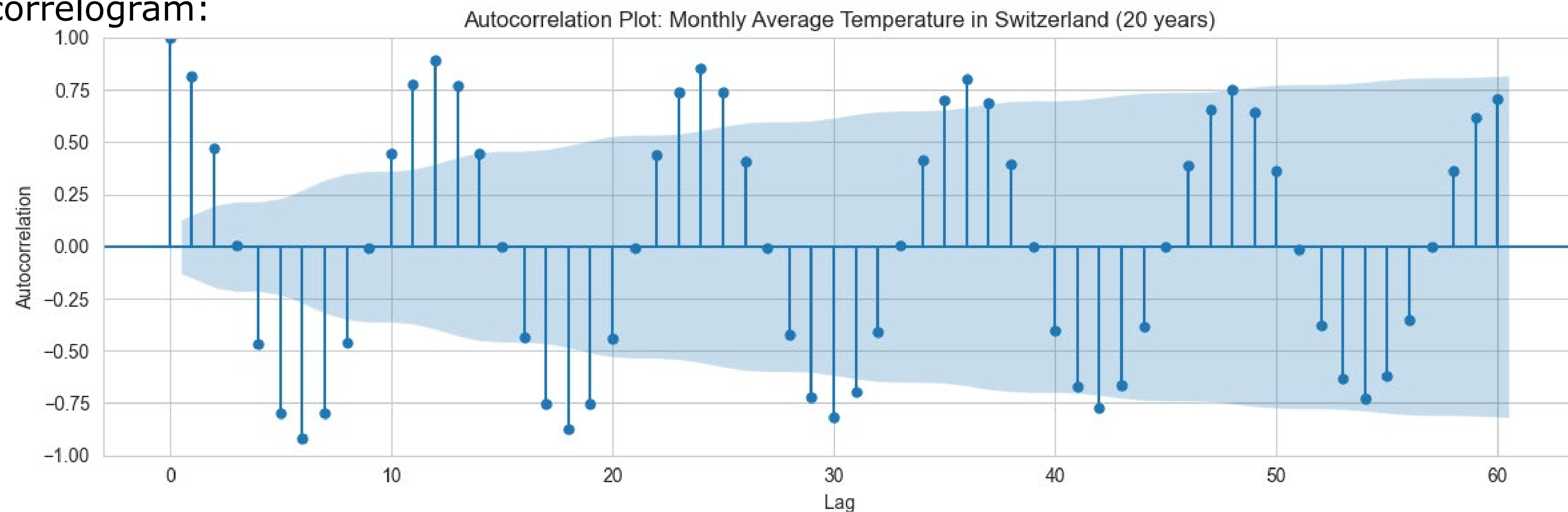
Higher Var X, High Var Y, Moderate Neg Corr
Covariance: -100.03



The auto-correlation function (ACF) $\rho(s, t)$



ACF plot or correlogram:



White noise (WN) process

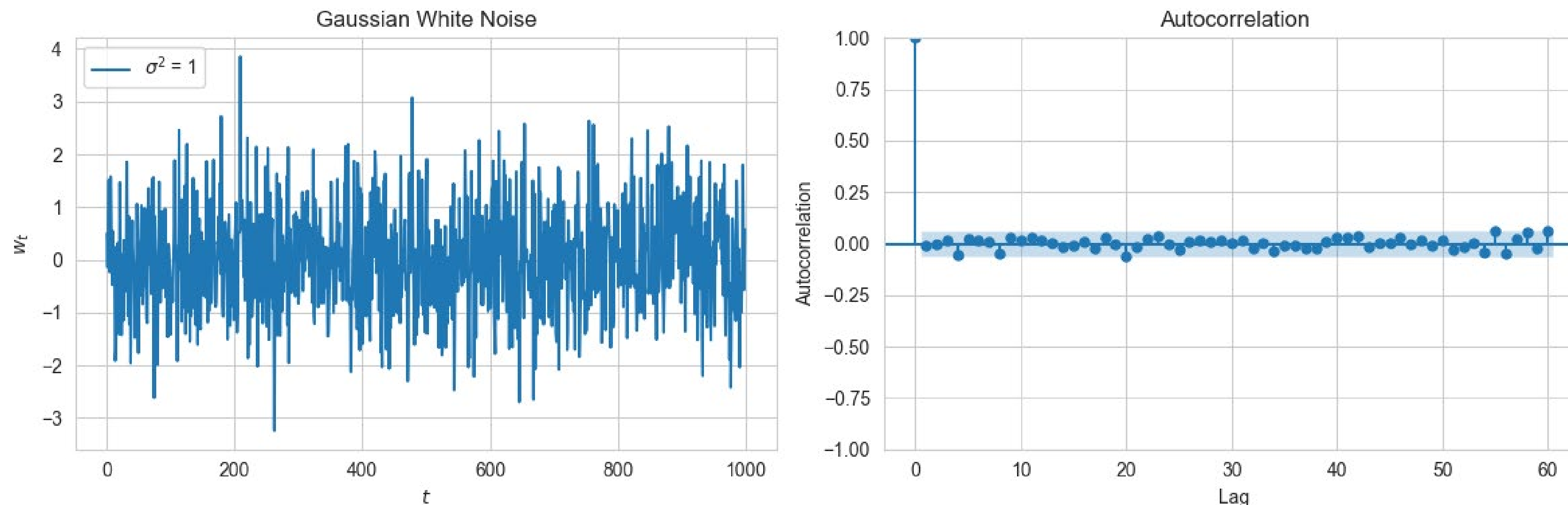
RVs $\{W_1, W_2, \dots, W_n\}$ are **uncorrelated** ($\text{Cov}(W_s, W_t) = 0, \forall s \neq t$), with **zero-mean** and **constant finite variance** σ^2 :

$$W_t \sim WN(0, \sigma^2)$$

An additional assumption is that the RVs are **independent** and **identically distributed** $W_t \sim iid WN(0, \sigma^2)$:

$$P[W_1 = w_1, \dots, W_n = w_n] = P[W_1 = w_1] \dots P[W_n = w_n]$$

Another useful assumption is that the RVs follow a **Gaussian** distribution (Gaussian WN) $W_t \sim \mathcal{N}(0, \sigma^2)$.



Signal + Noise model and time series decomposition

Signal + noise model: $X_t = \theta_t + E_t$ where θ_t is a modellable signal and E_t are errors (also called **innovations**).

Time series decomposition separates X_t into three components:

- Trend/trend-cycle U_t : increasing/decreasing pattern in the data.
- Seasonality S_t : repeating pattern with roughly fixed period.
- Remainder R_t : everything remaining (**residuals**), E_t if θ_t is fully represented by U_t and S_t .

Additive decomposition $X_t = U_t + S_t + R_t$

- Magnitude of trend and seasonal fluctuations do not vary with the value of the TS.

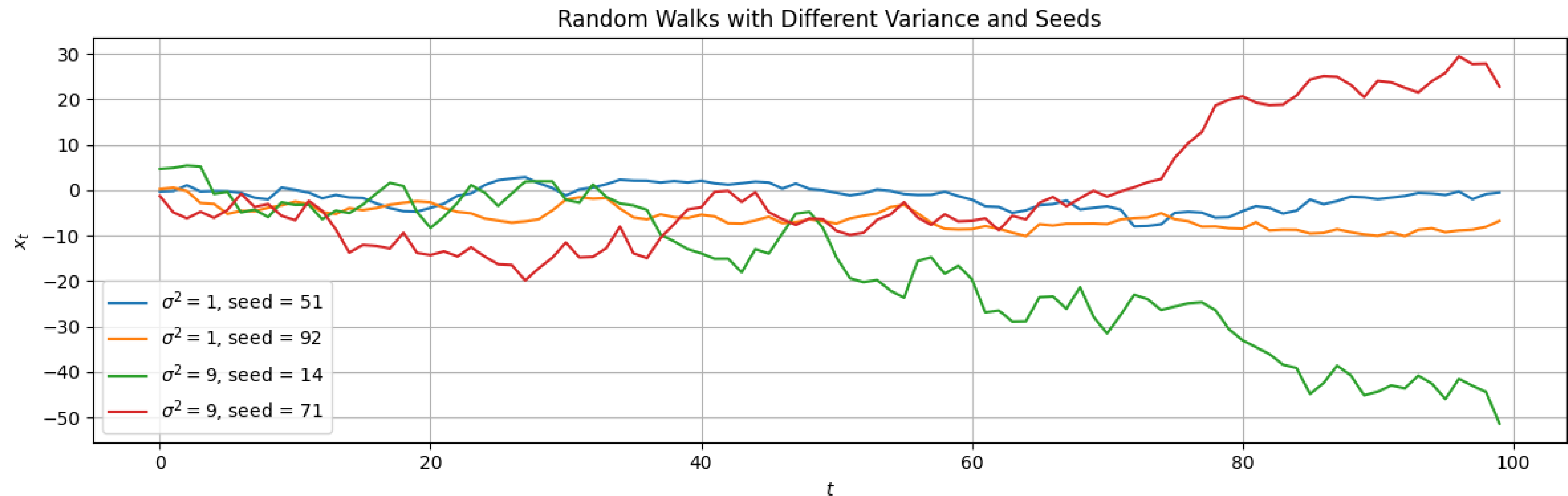
Multiplicative decomposition $X_t = U_t \times S_t \times R_t$

- Magnitude of trend and seasonal fluctuations are proportional to the value of the TS.
- Equivalent to the additive decomposition of the **log transformed** TS.

Model with noise – Random walk (RW) process

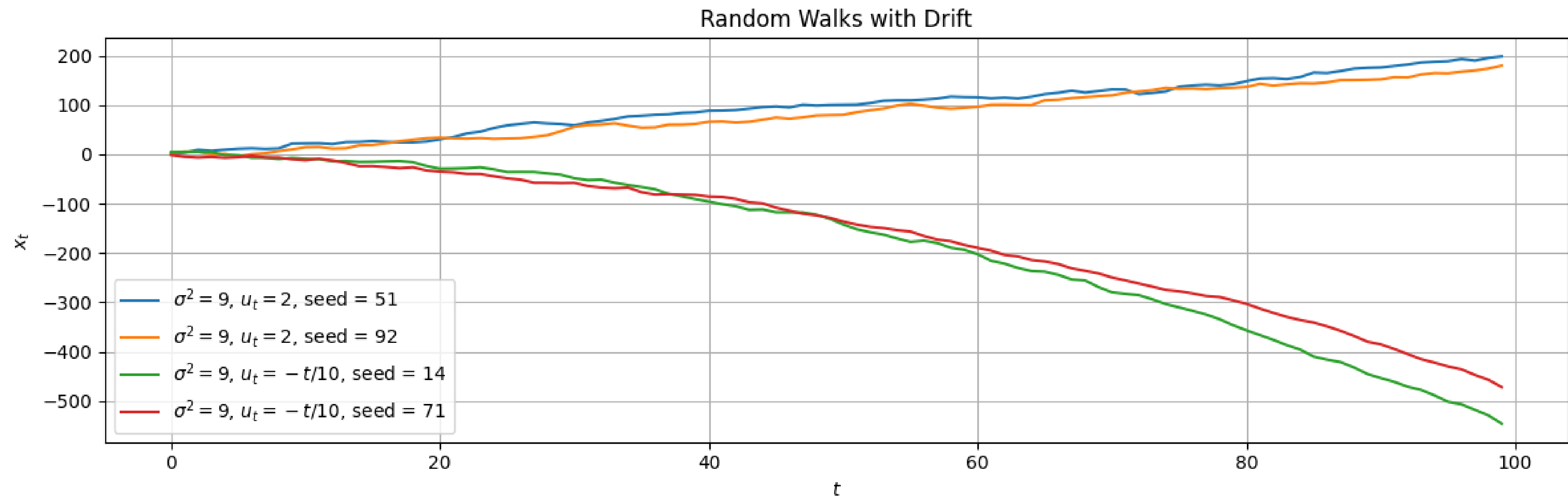
A **random walk** is a cumulative sum of iid zero-mean RVs: $X_t = \underbrace{0}_{U_t} + \underbrace{0}_{S_t} + \underbrace{\sum_{i=0}^t W_i}_{R_t}$ with $W_i \sim iid\ WN(0, \sigma^2)$

A RW process is **not iid** since its RVs are correlated: $X_t = X_{t-1} + W_t$ with $X_0 = W_0$.



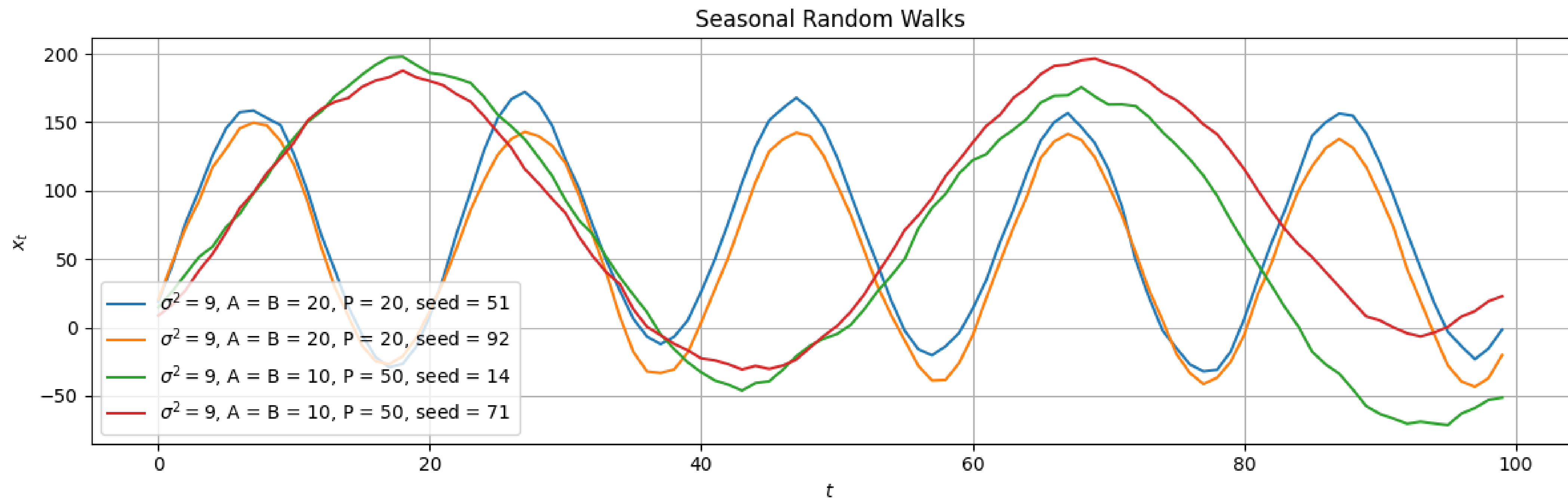
Model with trend – Random walk with drift

Random walk with drift: $X_t = X_{t-1} + W_t + U_t = \underbrace{\sum_{i=0}^t U_i}_{U_t} + \underbrace{0}_{S_t} + \underbrace{\sum_{i=0}^t W_i}_{R_t}$ with $X_0 = W_0$.



Model with seasonality – Seasonal random walk

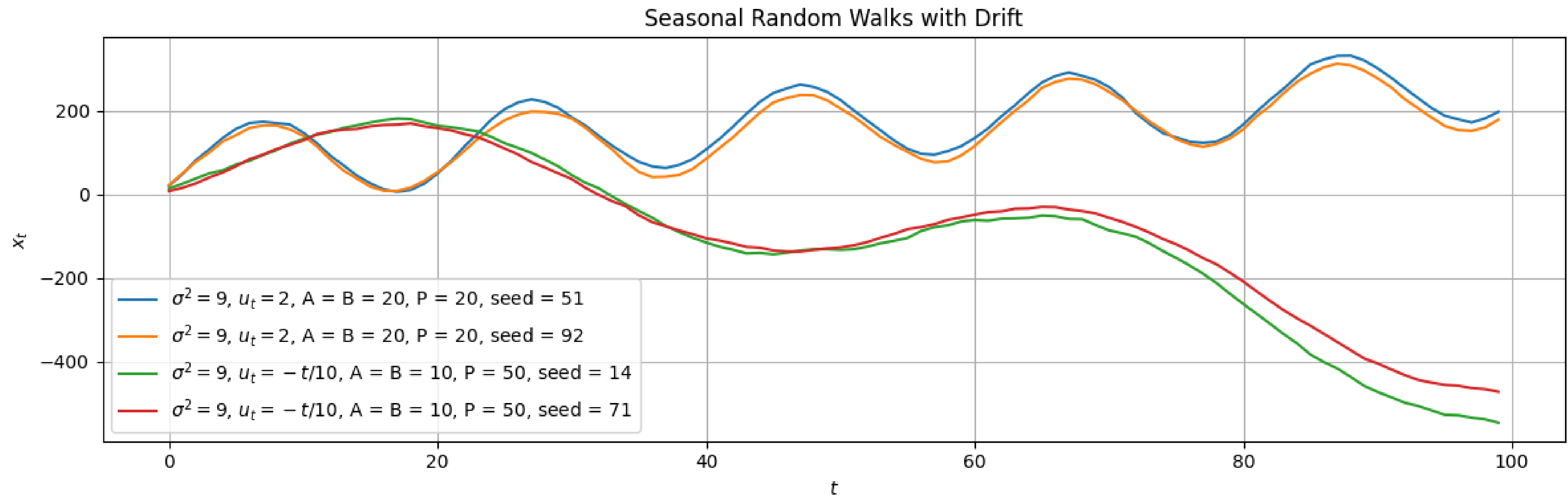
Seasonal random walk: $X_t = X_{t-1} + W_t + A \sin\left(\frac{2\pi t}{P}\right) + B \cos\left(\frac{2\pi t}{P}\right) = \underbrace{0}_{\tilde{U}_t} + \underbrace{\sum_{i=0}^t \left(A \sin\left(\frac{2\pi i}{P}\right) + B \cos\left(\frac{2\pi i}{P}\right) \right)}_{S_t} + \underbrace{\sum_{i=0}^t W_i}_{R_t}$ with $X_0 = W_0$



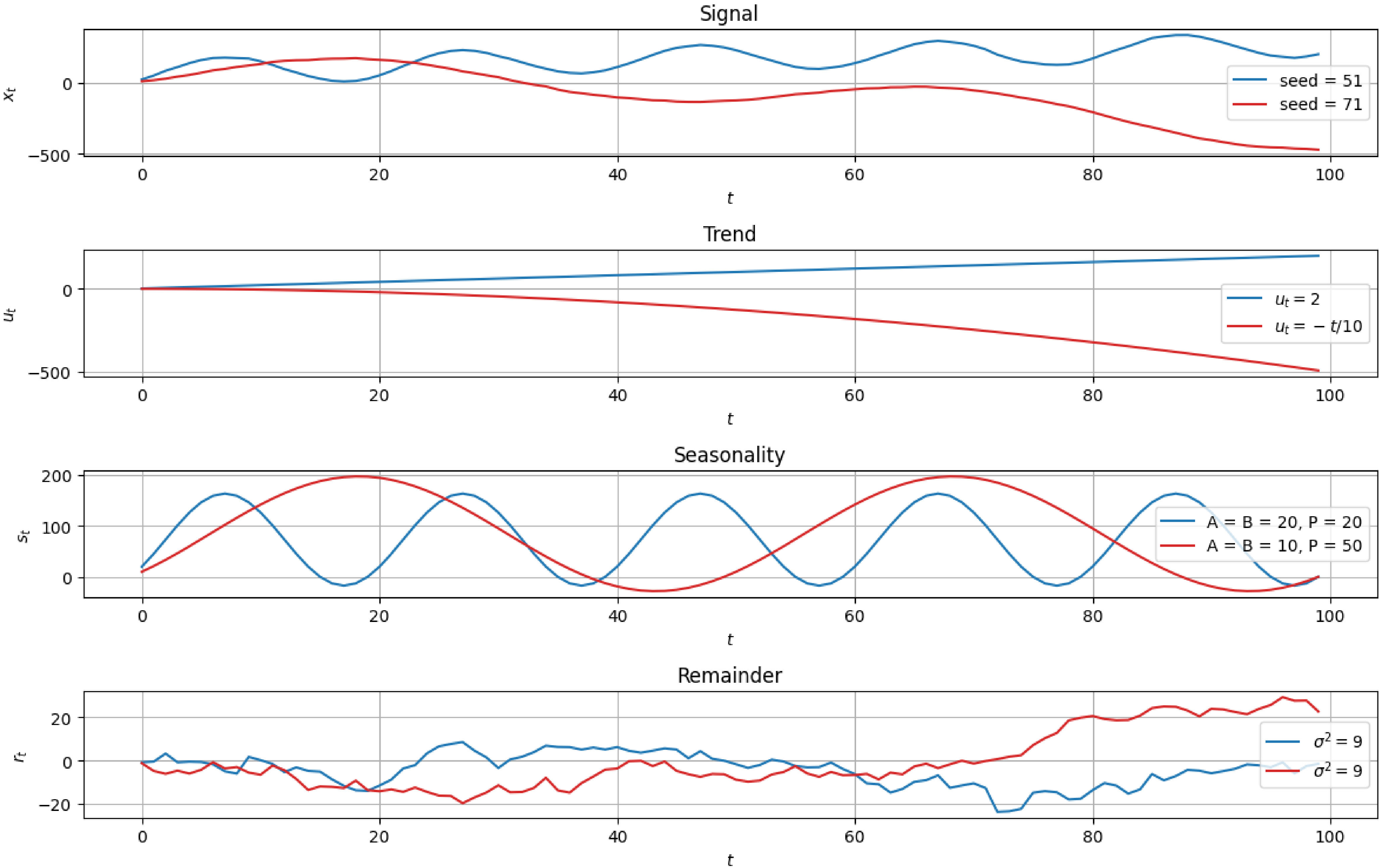
Model with trend and seasonality – Seasonal random walk with drift

Seasonal RW with drift:
$$X_t = X_{t-1} + W_t + U_t + A \sin\left(\frac{2\pi t}{P}\right) + B \cos\left(\frac{2\pi t}{P}\right) = \underbrace{\sum_{i=0}^t U_i}_{U_t} + \underbrace{\sum_{i=0}^t \left(A \sin\left(\frac{2\pi i}{P}\right) + B \cos\left(\frac{2\pi i}{P}\right)\right)}_{S_t} + \underbrace{\sum_{i=0}^t W_i}_{R_t}$$

with $X_0 = W_0$.



Decomposed Seasonal Random Walks with Drift



Decomposing time series with unknown components

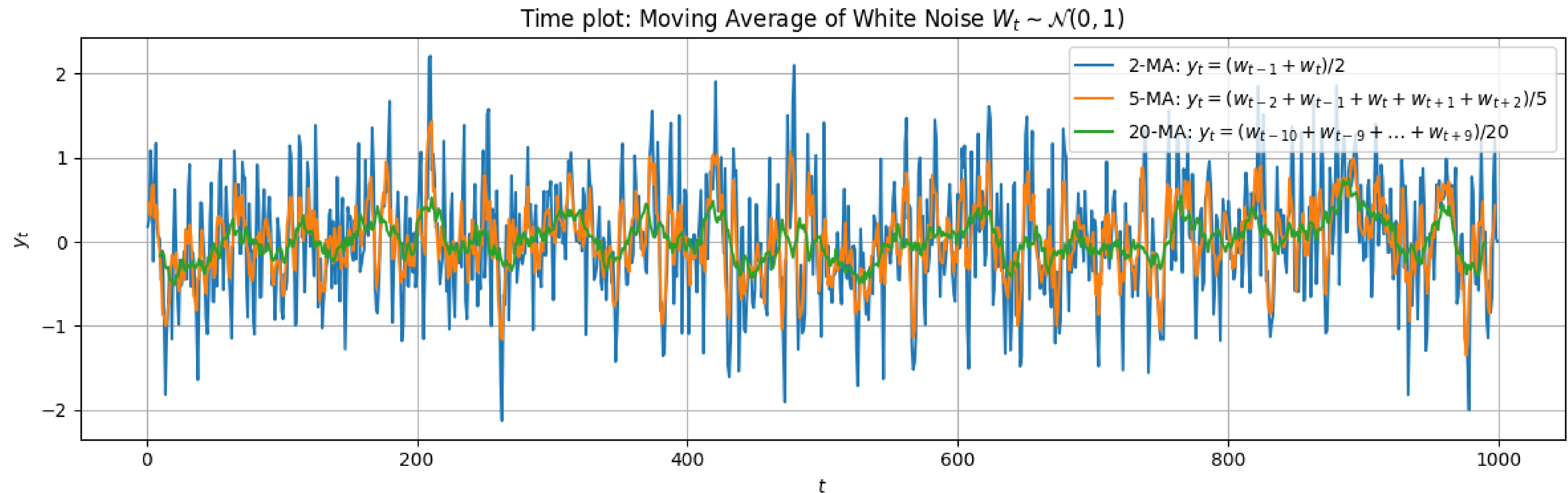
Given a time series realization $\{x_1, x_2, \dots, x_n\}$, assuming additive decomposition:

1. Estimate trend component $\hat{u}_t$
 - Moving average smoothing (classical decomposition)
 - LOESS – locally estimated scatterplot smoothing (STL decomposition)
2. **Detrend** time series $\hat{d}_t = x_t - \hat{u}_t$
3. Estimate seasonality component $\hat{s}_t$
 - Period adjusted averages (classical decomposition)
 - LOESS – locally estimated scatterplot smoothing (STL decomposition)
4. **Deseasonalize** TS to estimate the remainder $\hat{r}_t = \hat{d}_t - \hat{s}_t$

Estimating the trend – Moving average (MA) smoothing

Linear combinations of a TS values are called **linear filters**. The resulting TS is called a **filtered TS**.

A moving average of order m (m -MA) **smooths** a TS by averaging **m consecutive points**: $Y_t = \frac{1}{m} \sum_{i=-[(m-1)/2]}^{[(m-1)/2]} X_{t+i}$
(called window)



Estimating the trend – Moving average (MA) smoothing

Variations

- **Window center can be shifted** e.g., trailing 3-MA $Y_t = \frac{1}{3}(X_{t-2} + X_{t-1} + X_t)$
- Different weights can be assigned observations in the window (**weighted MA**) $Y_t = \frac{1}{m} \sum_{i=-[(m-1)/2]}^{[(m-1)/2]} a_i X_{t+i}$

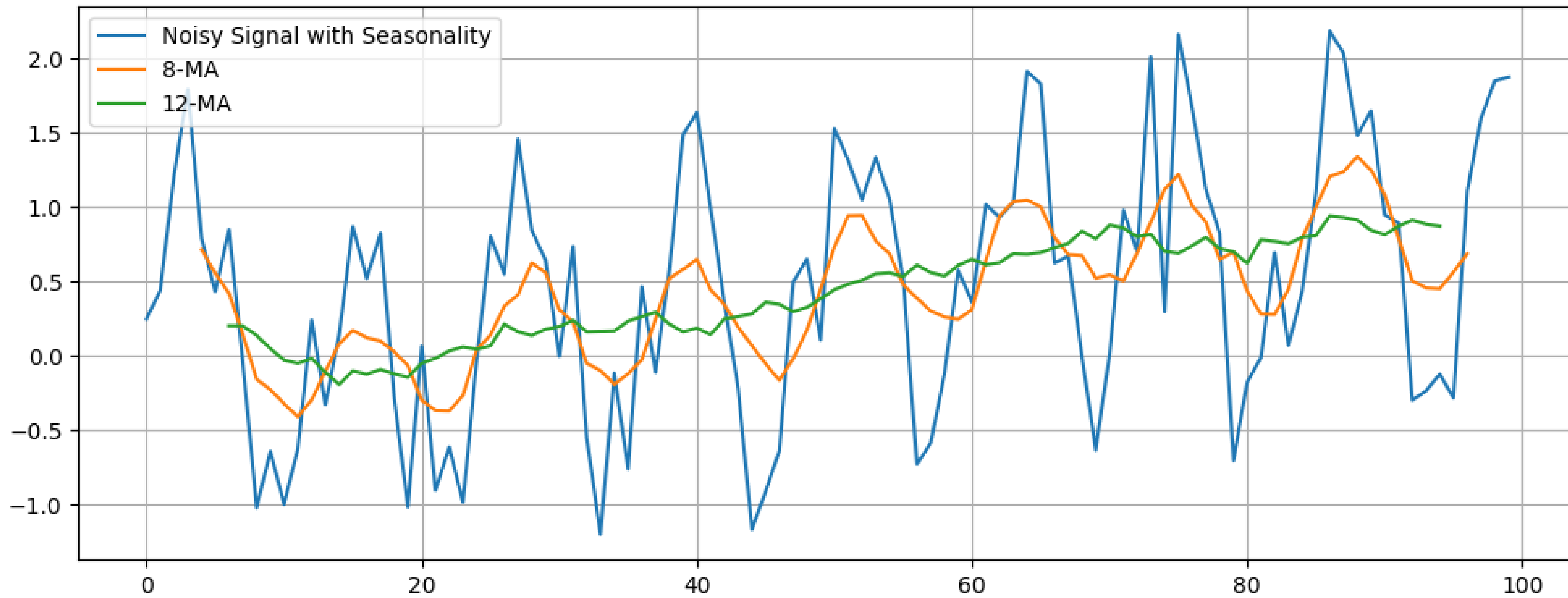
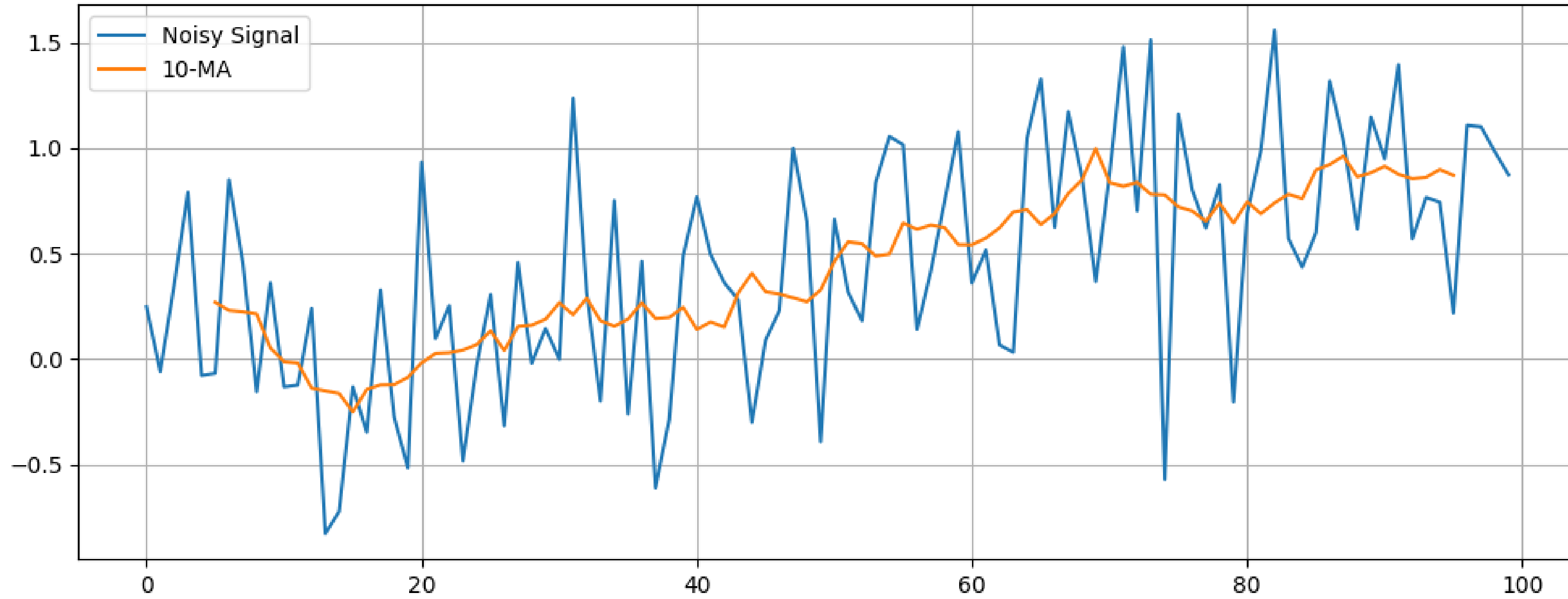
MA can be applied iteratively e.g., 4-MA then 2-MA referred as 2 x 4-MA.

- Additional smoothing
- Symmetry for even orders: 2 x m-MA is equivalent to a weighted (m+1)-MA with $w = \left[\frac{1}{2m}, \frac{1}{m}, \frac{1}{m}, \dots, \frac{1}{m}, \frac{1}{2m} \right]$

MA for trend estimation $\hat{u}_t = y_t$:

- Smooth out fluctuations to reveal underlying trends and cycles.
- Remove seasonal components to understand trend and cyclical behavior: **match order with period.**
- **Sensitive to outliers**

Trend Estimation with Moving Average



Estimating the seasonality – Period adjusted averages

Given a detrended TS realization $\{d_1, d_2, \dots, d_n\}$ with period P , assuming additive decomposition:

1. **Estimate** seasonal component $\bar{s}_t = \frac{1}{n} \sum_{i=0}^{\lfloor n/P \rfloor} d_{t+iP}$
2. **Adjust** seasonal components to ensure they sum to zero $\bar{\bar{s}}_t = \bar{s}_t - \frac{1}{P} \sum_{i=1}^P \bar{s}_i$
3. **Repeat** the seasonal component values for each period $\hat{s}_t = \bar{\bar{s}}_{t \bmod P}$

Seasonal and trend decomposition using LOESS (STL)

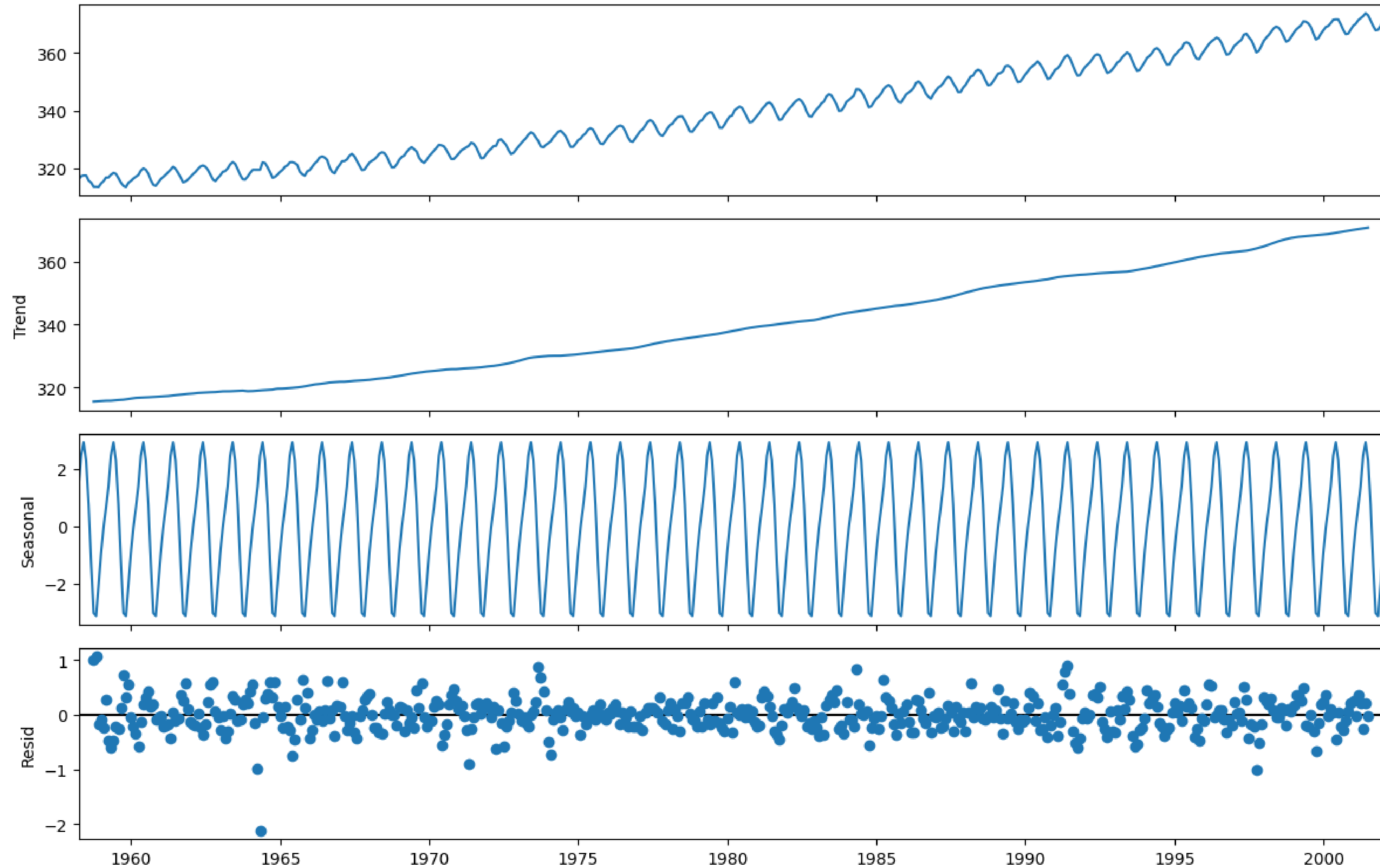
LOESS (locally estimated scatterplot smoothing) is a non-parametric regression method used to fit data.

- **Generalization** of moving average and polynomial regression able to capture non-linear trends.
- Apply a **sliding window** across the dataset, where at each point, a small subset of neighboring points is selected to fit a local linear/polynomial model.
- Neighbors are weighted according to their proximity.
- The size of the window (span) determines the **smoothing strength**.

STL uses LOESS to estimate both the trend and seasonality components

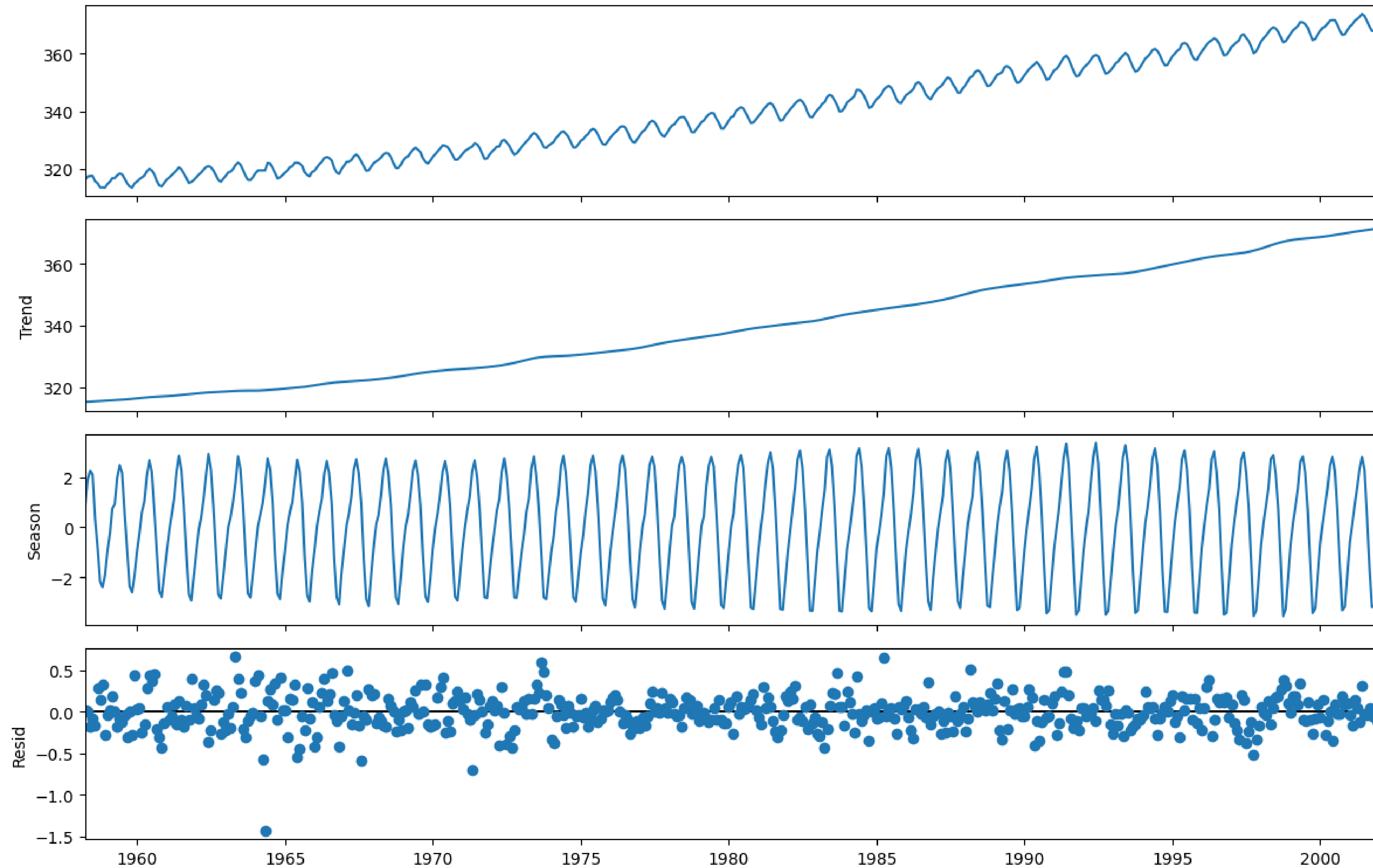
- Different sliding window spans for trend and seasonality
- In comparison to period adjusted averages, **seasonal component is allowed to change over time**.
- Can down-weight outliers to reduce their impact.

Classical decomposition: Mauna Loa Monthly Atmospheric CO2 Data

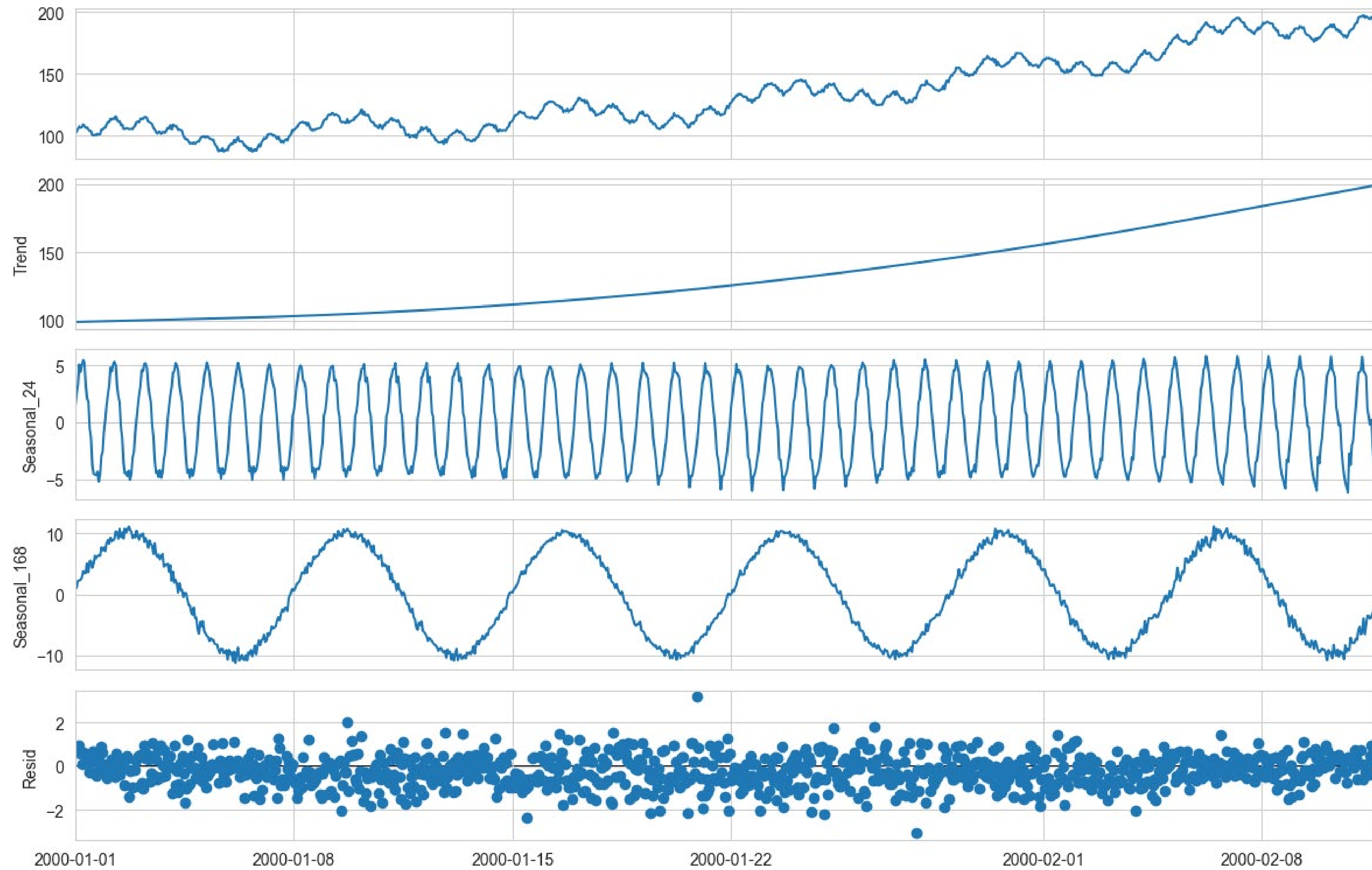


`statsmodels.tsa.seasonal.seasonal_decompose`

STL decomposition: Mauna Loa Monthly Atmospheric CO2 Data



Multiple Seasonal-Trend decomposition using LOESS



Exercise

Generate 2-3 **synthetic time series** with known components.

- Apply classical and STL time series decomposition.
- Review how well components are extracted, compute the **mean squared error**.

Generate and **interpret** the ACF plots of the different forms of random walks presented in this lecture.

Reimplement classical decomposition.

Extend previous exercise with ACF plots and time series decomposition.

- **Compare** classical and STL decomposition results.
- **Interpret** ACF plots and time series components.